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116th Doctoral Dissertation
Thesis Advisor : Kyong Ju Kim

**Development of an Automated Construction
Contract Review Framework Using LLM
and Domain Knowledge**

February 2025

**The Graduate School
Chung-Ang University
Department of Smart Cities
Major in Smart Planning, Construction & Information System
Eu Wang Kim**

Development of an Automated Construction Contract Review Framework Using LLM and Domain Knowledge

**Presented to the Faculties of the Chung-Ang University in
Partial Fulfillment of the Requirement of the Degree of
Doctor of Philosophy**

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Abstract

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Reviewing construction contracts requires expertise in both engineering and law, yet many organizations lack specialized professionals. While Large Language Models (LLMs) have potential, they often struggle to integrate domain-specific knowledge or produce reliable outputs for complex contracts. Addressing these challenges, this study introduces the Clause-Oriented Review with Agent-based Retrieval-Augmented Generation (CORA-RAG) system, which enhances analysis accuracy and reliability through three key components: (1) integrating structured domain knowledge via a knowledge-enhanced RAG pipeline aligned with Standard Contract Forms (SCFs); (2) employing classification-based filtering combined with hybrid dense-sparse retrieval mechanisms to improve retrieval accuracy; and (3) utilizing a multi-agent LLM framework with a multi-stage generation process that includes context refinement and agent collaboration.

CORA-RAG's knowledge base comprises over 2,000 text segments

mapped to SCFs, enabling precise interpretation of contractual provisions. The system uses a multi-agent approach, with each agent LLM focusing on different aspects of analysis, such as semantic understanding, context analysis, and risk assessment. The structured analysis progresses through stages: detecting differences, refining context, addressing practical and legal aspects, and conducting detailed risk and opportunity assessments—all facilitated by agent collaboration in a comprehensive generation framework.

Validation involved quantitative evaluation using Retrieval-Augmented Generation Answering System (RAGAS) metrics and expert assessments. Using Oman's General Conditions as a case study, performance evaluations showed significant improvements: a 52.6% increase in Context Recall and a 48.3% increase in Faithfulness compared to basic RAG approaches, particularly in complex reasoning and multi-context understanding. Classification-based filtering combined with hybrid retrieval methods reduced irrelevant data early in the process, while the multi-agent LLM framework enhanced the reliability and accuracy of content generation.

Expert evaluations from thirteen construction industry professionals, including contractors, clients, and consultants, indicated that the system operates at an analytical level equivalent to junior professionals with 2–4 years of experience. Experts confirmed the system's utility in expediting

basic contract analysis and enhancing work efficiency by reducing review time. It effectively assists practitioners, especially those with less experience, in navigating complex contracts by identifying relevant clauses and highlighting deviations. However, limitations were noted in interpreting nuanced language and conducting sophisticated legal analyses and risk assessments—areas typically requiring more experienced professionals.

Key contributions of this study include developing an SCF-based structured knowledge framework, enhancing retrieval accuracy through classification-based filtering combined with hybrid dense-sparse retrieval techniques, and implementing a context-aware, multi-stage answer generation process via a multi-agent LLM framework. By integrating structured knowledge with advanced agent collaboration in both retrieval and generation, the CORA-RAG system provides construction professionals with a reliable and efficient tool for detailed contract reviews, addressing the growing need for automated expertise in construction contract analysis.

Keywords: AI-driven Review, Contract Management, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Knowledge Base

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1. Introduction

1.1 Background and Motivation

Construction projects often encounter disputes due to inadequate contract management, which includes ambiguous terms, poor communication, and misinterpretations of obligations. Errors and omissions in contract documents are primary causes of disputes in major construction markets, significantly impacting project success by escalating costs and delaying timelines (Arcadis, 2022). Recent industry reports have highlighted a dramatic rise in construction dispute costs over a short period (Arcadis, 2023).

To address these challenges, improving the efficiency of contract management is essential. The emergence of Artificial Intelligence (AI) and Large Language Models (LLMs) offers promising approaches to provide expert knowledge and increase productivity. Specifically, AI-driven contract analysis tools have the potential to streamline contract review processes and strengthen risk management capabilities. By automating repetitive tasks, these technologies enable construction professionals to manage contracts more efficiently.

However, applying current LLMs in contract management presents

limitations. As general-purpose models, they raise concerns about the trustworthiness and accuracy of their outcomes when applied to precision-critical contractual contexts. The lack of domain-specific knowledge can lead to missing subtle meanings in complex clauses, hindering the models' ability to interpret contracts holistically. This difficulty makes it challenging to provide reliable analysis across entire documents, as nuances and interdependencies between clauses may be overlooked or misinterpreted.

1.2 Research Objectives

This research aims to overcome the significant challenges in automating construction contract reviews by introducing a structured knowledge-integrated Retrieval-Augmented Generation (RAG) pipeline tailored for construction contracts. The proposed system, named Clause-Oriented Review with Agentic RAG (CORA-RAG), integrates domain expertise with the advanced capabilities of language models. The study focuses on three primary objectives:

1. Develop a Structured Knowledge Framework: Systematically organize technical and legal expertise by mapping contract clauses to relevant analyses and practical considerations. This framework ensures

comprehensive coverage of essential aspects in contract review and enables the system to interpret complex contractual relationships with clarity and precision. It utilizes Standard Contract Forms (SCFs) such as FIDIC and Oman's General Conditions.

2. Create a Robust Retrieval Mechanism: Maintain accuracy when dealing with complex and extensive contract data by employing a two-stage retrieval process. Initially, classification-based filtering narrows the search scope within the knowledge base, enhancing focus and reducing irrelevant data. Subsequently, a hybrid search approach combines dense retrieval (semantic matching) and sparse retrieval (keyword-based matching) to ensure both contextual understanding and precise identification of critical contractual elements.

3. Establish Reliable Generation Processes: Integrate structured references into the system's outputs to enhance response accuracy and reliability. By refining retrieved contexts and incorporating domain-specific knowledge, the system produces analysis outputs that align with legal, technical, and practical requirements, ensuring that generated insights are not only accurate but also actionable in professional settings.

Through these objectives, the proposed CORA-RAG system aims to

enhance the efficiency of contract reviews by addressing key challenges such as the lack of domain expertise, maintaining accuracy in analysis, and ensuring reliable results. The CORA-RAG system follows a step-by-step process: it begins by identifying differences between new contracts and standard forms, refines the context by analyzing interpretation, practical issues, and legal aspects, and finally performs a detailed review—including risk and opportunity assessment—to ensure a thorough understanding of the contract. By combining structured knowledge with advanced LLM technologies, CORA-RAG bridges the gap between traditional manual methods and modern automated tools, enhancing the practicality and reliability of construction contract reviews.

1.3 Thesis Structure

This thesis is organized into six chapters to systematically present the research background, methodology, findings, and contributions.

Chapter 1: Introduction provides an overview of the research, detailing the background, motivation, and objectives. It underscores the challenges faced in construction contract review and introduces the CORA-RAG system as a potential solution to these issues.

Chapter 2: Research Background reviews the existing challenges in construction contract management and explores the applications of AI and LLM technologies. It identifies key research gaps, such as the need for structured knowledge integration, effective retrieval mechanisms, and reliable output generation.

Chapter 3: Retrieve-Focused Framework and Validation describes the architecture of the CORA-RAG system's retrieval-focused framework. It explains the design of the knowledge base, retrieval mechanisms, and system components, followed by an experimental validation to assess retrieval performance.

Chapter 4: Generate-Focused Framework and Validation focuses on the generation aspect of the system, detailing the multi-stage generation process, context refinement pipeline, and system implementation. This chapter also presents expert evaluations to validate the quality and reliability of the generated outputs.

Chapter 5: Discussion provides an in-depth analysis of the system's technical performance and practical implications. It discusses the benefits and applications of the system while addressing its current limitations and challenges.

Chapter 6: Conclusion summarizes the key findings and contributions of the research. It also outlines the study's limitations and offers recommendations for future research.

2. Research Background

2.1 Challenges in Construction Contract Review

The importance of construction contract review has grown significantly as the value of disputes escalates. This process requires expert knowledge that is both timely and appropriately scoped. Engineers frequently encounter challenges with legal implications, while lawyers may struggle to grasp the technical details (Seppälä, 2023, p. 162). The complexity of international projects is further compounded by the need to interpret contracts across various legal systems, necessitating a thorough understanding of diverse contractual forms and legal principles (Dmaidi et al., 2013).

Expertise in construction contract review involves understanding, analyzing, and interpreting complex contract provisions that intertwine technical and legal aspects. Professionals must possess deep knowledge of construction processes, contract law, industry practices, and specific project contexts. This integration is essential to identify risks, provide clear interpretations, and align with project objectives and the parties' intentions. The complexity is notably higher in FIDIC contracts, where provisions across different clauses are highly interconnected. FIDIC contracts serve as

fundamental building blocks in construction project delivery worldwide, leading to extensive studies and applications (Migliaccio & Shrestha, 2022, p. 283).

Understanding these contracts demands a comprehensive approach to analyze how various clauses interact (Baker et al., 2013, p. 30). The presence of numerous cross-references and complex sentence structures increases the risk of misunderstanding (Rasslan & Nassar, 2017). Contract interpretation principles require reviewers to adopt a holistic approach rather than focusing on isolated words or phrases, necessitating a thorough comprehension of the entire document (McMeel, 2011, p. 80).

Table 1 illustrates the progressive expansion of FIDIC's contract suite from a single contract form in 1977 to 18 distinct works contracts by 2023. This expansion reflects both the growing complexity of the construction industry and the increasing challenges faced by contract review professionals. The suite's growth shows a strategic pattern of diversification, from the traditional Red Book to specialized forms like the Gold Book for design-build-operate projects (2008) and the Emerald Book for underground works (2019). Each new addition targets specific project delivery methods or specialized construction sectors, significantly expanding the knowledge requirements for professionals.

This expansion has created two major challenges: professionals must now maintain expertise across a wider range of contract types, each with specific provisions, and understand the complex interrelationships between different contract forms. The periodic updates and reprints, including the 2022 reprints of the 2017 Rainbow Suite, further necessitate continuous learning to maintain current expertise across this expanding contract portfolio.

Table 1. Progressive Expansion of FIDIC Contract Suite (1977-2023)

Year	Cumulative Count	Works Contract
1977	1	Red Book (3rd Edition)
1987	3	Red Book (4th Edition), Yellow Book (3rd Edition)
1995	4	Orange Book (1st Edition)
1999	8	Red Book (1st Edition), Yellow Book (1st Edition), Silver Book (1st Edition), Green Book (1st Edition)
2005	9	Pink Book (MDB Harmonised Edition, Version 1)
2006	11	Pink Book (MDB Harmonised Edition, Version 2), Blue-Green Book (1st Edition)
2008	12	Gold Book (1st Edition)
2010	12	Pink Book (MDB Harmonised Edition, Version 3)
2016	13	Blue-Green Book (2nd Edition)
2017	16	Red Book (2nd Edition), Yellow Book (2nd Edition), Silver Book (2nd Edition)
2019	17	Emerald Book (1st Edition)
2021	18	Green Book (2nd Edition)
-	-	2022: Red Book, Yellow Book, Silver Book (2017 Editions) Reprint
-	-	2023: Emerald Book Reprint

2.2 AI and LLM Applications in Contract Management

Automated review in construction contract management is evolving, presenting both technological advancements and implementation challenges. Early research prominently featured rule-based Natural Language Processing (NLP) approaches for risk detection, compliance checking, and clause analysis (Hamie & Abdul-Malak, 2018; Lee et al., 2019; Serag et al., 2010; Xu & Cai, 2018; Zhang et al., 2016). These methods were later extended to include uncertainty analysis and automated compliance checking (Faraji et al., 2021; Guo et al., 2021).

Conventional machine learning methods began with litigation outcome prediction using various classifiers (Arditi & Pulket, 2005). Subsequent studies implemented Support Vector Machines (SVM), Naïve Bayes, and other machine learning approaches for compliance checking and requirements classification (Candaş & Tokdemir, 2022; Hassan et al., 2020; Salama & El-Gohary, 2016). Hybrid approaches combining rule-based and machine learning methods were also explored (Choi et al., 2021; Hassan & Le, 2020).

Deep learning and transformer-based models have further enhanced contract analysis capabilities. Models like Bidirectional Long Short-Term

Memory (Bi-LSTM), Conditional Random Fields (CRF), and Bidirectional Encoder Representations from Transformers (BERT) have been utilized to improve performance in tasks such as clause classification and information extraction (Moon et al., 2022; Xue et al., 2022; Zheng et al., 2022; Zhou et al., 2023). Recent studies have advanced the field using architectures like DeBERTa and pre-trained BERT models for risk prediction and dispute classification (Ko et al., 2024; Pham & Han, 2023; Qi et al., 2024; Zhong et al., 2024).

Recently, LLMs with knowledge-augmented approaches have emerged in contract management (Wong et al., 2023; Zheng et al., 2023). While these advancements offer significant potential, current methods still face substantial challenges:

- **Insufficient Domain Expertise:** LLMs, being trained on general datasets, often lack the specialized technical and legal knowledge required for in-depth contract analysis.
- **Performance Degradation with Lengthy Documents:** LLMs' accuracy notably drops when processing lengthy documents, with reported accuracy decreasing to 55% for documents exceeding ten paragraphs (Wang et al., 2024).

- **Reliability Concerns:** The probabilistic nature of LLM outputs leads to issues like hallucinations and inconsistent context understanding, which are critical in precision-sensitive contractual contexts.

These limitations are particularly evident in RAG systems, which continue to struggle with managing large volumes of unstructured information and maintaining consistent interpretation of complex contract relationships. This underscores the need for an improved methodology that effectively integrates domain expertise while ensuring reliable contract analysis capabilities.

2.3 Related Work and Research Gaps

The evolution from basic NLP techniques to advanced LLMs in AI-driven contract management highlights the increasing importance of integrating domain-specific knowledge with structured information to enhance analysis performance. Effective contract review requires extensive expertise in both technical and legal domains, making knowledge integration particularly challenging. This reflects the principle that "knowledge is power" in expert systems, where system performance is more heavily reliant

on domain knowledge quality than on reasoning capabilities (Brock, 2018).

A fundamental limitation of LLMs is their struggle with domain-specific knowledge that is not widely represented in their training data (Mallen et al., 2022). Kandpal et al. (2022) indicate that simply increasing model size does not significantly improve the recall of specialized or rare information. In the context of construction contracts, where specialized technical and legal expertise is essential, these limitations become especially significant.

Several critical challenges have been identified:

- **Difficulty in Organizing Domain Expertise:** Construction contracts require the integration of both technical and legal knowledge to analyze specific provisions. Current systems struggle to incorporate and maintain such diverse expertise in a structured format.
- **Accuracy Degradation with Complex Documents:** Managing lengthy and complex documents remains problematic. Existing basic RAG approaches, which handle unstructured data, face limitations in effective information retrieval and matching, leading to decreased accuracy.

- Reliability Issues in LLM-Generated Content: Ensuring result reliability is a critical concern. LLM-based systems are prone to hallucinations and may misinterpret complex contractual relationships, necessitating sophisticated verification mechanisms to ensure accuracy and consistency (Zheng et al., 2023).

Addressing these challenges requires an approach that effectively integrates structured domain knowledge into the AI models. This research proposes the development of a structured knowledge-integrated RAG pipeline that enhances the RAG process for construction contract review. The proposed solution focuses on:

- Implementing a Structured Knowledge Framework: Systematically organizing technical and legal expertise to provide a comprehensive knowledge base for analysis.
- Developing a Precise Retrieval Mechanism: Employing a two-stage retrieval process that combines classification-based filtering with dense-sparse retrieval methods to maintain accuracy when dealing with complex and extensive contract data.

- Establishing Reliable Answer Generation Processes:
Integrating structured references into the system's outputs
to improve response accuracy and reliability.

By bridging the gap between the challenges in practical contract review and the limitations of existing AI methods, this approach aims to enhance the practical applicability of automated contract review tools in construction projects. The integration of structured domain knowledge is expected to improve the accuracy, consistency, and reliability of AI-driven contract analysis, making it a valuable tool for professionals in the industry.

Table 2. Literature review

Authors	Year	Rule-based NLP	Conventional ML	DL/ Transformer	LLM	Research Purpose	Data Used	Features
Youssef et al.	2018	-	-	-	-	Contract Risk Identification	Construction Contracts	AHP
Ganbat et al.	2020	-	-	-	-	Contract Risk Identification	Survey Data	Social Network Analysis (SNA)
Zhang & El-Gohary	2016	O	-	-	-	Compliance checking	Construction Regulatory Documents	Rule-based NLP, Domain-specific ontology
Hamie & Abdul-Malak	2018	O	-	-	-	Classification of Priority Clauses	Construction Contract Documents	Rule-based NLP
Xu & Cai	2018	O	-	-	-	Compliance checking for the AEC domain	Regulatory Documents	Rule-based NLP, Semantic Frames, FrameNet, Semantic Role Labeling
Lee et al.	2019	O	-	-	-	Risk Factor Identification	Construction Contracts	Rule-based NLP, Parsey McParseface
Lee et al.	2020	O	-	-	-	Missing Clause Identification	Contract Text Data	Rule-based NLP
Faraji et al.	2021	O	-	-	-	Identification of Contract Obligations	Construction Contracts	Rule-based NLP, Lexicon Database
Guo et	2021	O	-	-	-	Compliance Checking for	Regulatory	SPARQL, BimSPARQL,

Authors	Year	Rule-based NLP	Conventional ML	DL/ Transformer	LLM	Research Purpose	Data Used	Features
al.						Construction Projects	Documents and BIM Data	ifcOWL
Serag et al.	2010	O	△	-	-	Contract Risk Identification	Construction Contract Clauses	Named Entity Recognition, Logic Reasoners
Salama & El-Gohary	2016	-	O	-	-	Support Automated Compliance Checking through Contract Document Classification	Construction Contracts	SVM, Naïve Bayes, Maximum Entropy, Deontic Model
Candaş & Tokdemir	2022	-	O	-	-	Requirements Identification	Construction Contracts	SVC, REG, DTC, GNB, MNB, MLP
Arditi & Pulket	2005	O	O	-	-	Judgment Prediction	Case Law Data	Boosted Decision Trees, ANN, ADABOOST
Hassan & Le	2020	O	O	-	-	Requirements Identification	Contract Documents	Word2Vec, Naïve Bayes, SVM, Neural Network
Hassan et al.	2020	O	O	-	-	Contract Requirements Classification	Construction Contracts	Naïve Bayes, SVM
Choi et al.	2021	O	O	-	-	Contract Risk Identification	ITB Documents	Random Forest, NER, phrase-matcher
Lee et al.	2023	O	O	-	-	Specification Review	Specification	Rule-based NLP,

Authors	Year	Rule-based NLP	Conventional ML	DL/ Transformer	LLM	Research Purpose	Data Used	Features
							Documents	Word2Vec, DSSM
Moon et al.	2022	-	-	O	-	Specification Review	Specification Documents	Word2Vec, Bi-LSTM, CRF, Doc2Vec
Xue et al.	2022	-	-	O	-	Contract Summarization/Interpretation	Construction Contracts	Distilbart, Pegasus, BART
Zheng et al.	2022	-	-	O	-	Complex Regulation Interpretation	Web Crawling Data	BERT
Zhou et al.	2023	-	-	O	-	Missing Clause Identification	Construction Contracts	BERT, MatchPyramid, CNN, LSTM
Qi et al.	2024	-	-	O	-	Contract Risk Identification	Construction Contracts	DeBERTa
Pham & Han	2023	-	O	O	-	Risk and Responsibility Prediction	Construction Contracts	BERT, BiLSTM, CNN
Zhong et al.	2024	-	O	O	-	Contract Risk Identification	Case Law Data	TextCNN, LSTM, RCNN, Transformer
Zheng et al.	2023	-	-	O	O	Complex Regulation Interpretation	Regulatory Texts	LLM-FuncMapper, BERT
Wong et al.	2023	O	O	O	O	Contract Risk Identification	Contract Documents	GPT, DeBERTa, BART, SVM, Naive Bayes

3. Retrieve-Focused Framework and Validation

3.1 System Requirements

To effectively address the challenges in automating construction contract reviews, the proposed system must meet several key requirements. These requirements focus on integrating expert knowledge, incorporating precise retrieval mechanisms, generating accurate and traceable outputs, and maintaining connections to reference sources—crucial for processing complex construction contracts.

First, the system needs to integrate technical and legal expertise through a systematically organized knowledge structure. LLMs often struggle with domain-specific knowledge that isn't widely represented in their training data, especially in specialized fields like construction contracts. Therefore, integrating structured knowledge into the RAG pipeline is essential for accurately analyzing specific clauses and handling specialized terms. This ensures comprehensive coverage and precise interpretation of contract clauses.

Second, the system must incorporate precise retrieval mechanisms to effectively extract relevant clauses and information from extensive text documents. To overcome limitations in handling large volumes

of unstructured information and maintaining consistent interpretations, the system should employ a two-stage retrieval process. This includes classification-based filtering to narrow down pertinent sections and dense-sparse retrieval methods to extract precise information, even from lengthy and complex documents with interconnected clauses.

Third, the system must generate answers that are not only accurate but also traceable to their sources. Given the uncertainty inherent in LLM outputs, implementing reference-based answer generation ensures that all responses are supported by and linked to specific sections of the knowledge base. This approach enhances the reliability and trustworthiness of the automated analysis, allowing users to verify outputs and maintain confidence in the system's recommendations.

3.2 Overview of Retrieve-Focused System Architecture

This research proposes an approach that addresses these requirements:

- Implementing a Structured Knowledge Framework that organizes technical and legal expertise.
- Developing a Precise Retrieval Mechanism that maintains accuracy even with complex documents.
- Establishing Reliable Answer Generation Processes based on structured reference knowledge.

The proposed framework enhances the conventional RAG pipeline by integrating structured domain knowledge with a two-stage retrieval process and reference-based generation, ensuring reliability and precision in contract analysis.

The system's core functionality includes three main components:

1. Knowledge Base: A repository containing structured contract information. It maps clauses to relevant analyses, interpretations, and practical considerations. This structured knowledge framework ensures comprehensive coverage and precise interpretation of contract clauses by

leveraging SCFs.

2. Enhanced Retrieval Mechanism: A retrieval component that combines classification-based filtering with dense-sparse search methods to improve the accuracy of retrieving relevant information. This hybrid approach effectively narrows down pertinent sections and precisely extracts information, even from lengthy and complex documents with interconnected clauses.

3. Generation Component: This component performs differential analysis by comparing the input contracts to SCFs while ensuring reliability through contextual interpretation. It analyzes identified differences using expert analysis, legal information, practical issues, and suggested improvements related to the matched SCF clauses.

These components are coordinated by a System Controller that manages user interactions and integrates the components to ensure proper data flow and validation. While the core processing flow is sequential through these components, the system controller facilitates interactions among components to maintain efficiency and accuracy.

Figure 1 illustrates the system architecture and process flow. When a user inputs a new project contract, it is processed through the three main

components:

- The Knowledge Base stores structured expert knowledge and contract information.
- The Retrieval Component searches and matches relevant information using the hybrid retrieval mechanism, identifying modified, added, and omitted clauses by comparing the input contract against SCFs, ensuring both precision and relevance.
- The Generation Component analyzes these differences, providing comprehensive and reliable analysis outputs.

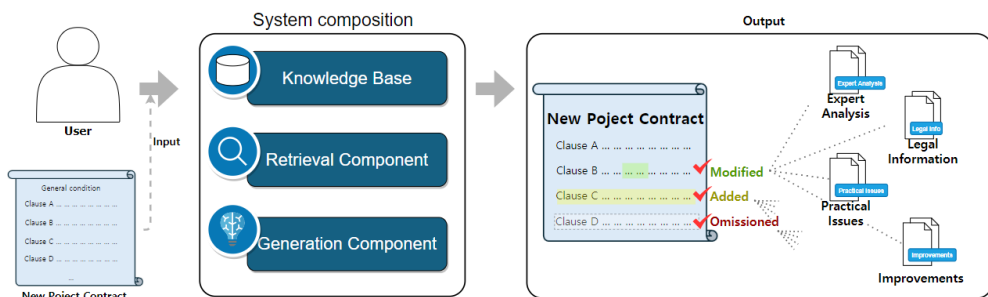


Figure 1. System architecture

Figure 2 presents an architectural comparison between the proposed CORA-RAG methodology and existing approaches. The right side of the

diagram depicts conventional Standard RAG and Search-Augmented Generation methods, which typically involve simple combinations of a knowledge base with a base LLM or a web search engine with a base LLM, respectively. In contrast, the proposed CORA-RAG method, shown on the left, features a more sophisticated structure with the three main components described above. This comparison highlights how CORA-RAG overcomes the limitations of existing approaches, enabling more accurate and reliable contract analysis by integrating structured domain knowledge, advanced retrieval mechanisms, and multi-stage analysis processes.

Table 3 compares information retrieval and context processing methods across different approaches, underscoring the distinctive features of the proposed method.

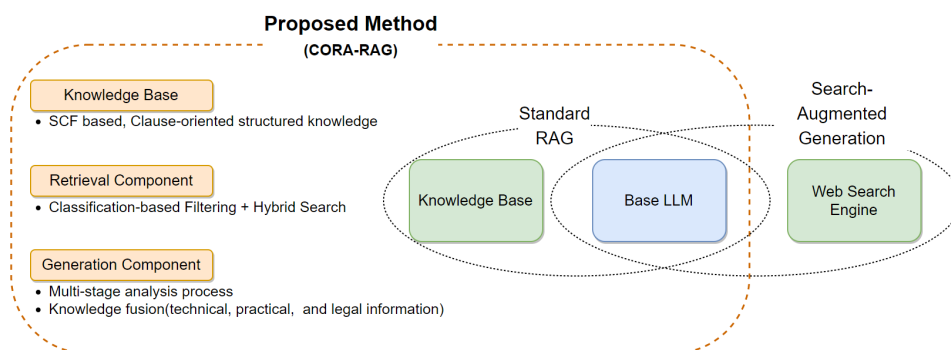


Figure 2. Comparison of CORA-RAG with Existing Approaches

Table 3. Comparison of LLM approaches

	Information Retrieval Mechanism	Context Information	Prompt	Limitations
Base LLM	None	None	-Basic user prompt only	- Hallucination issues - High cost requirements (for fine-tuning)
Search-Augmented LLM	Web search	- Unstructured web information - Combined expert and non-expert information - Dynamic online content	- User prompt - Retrieved web context	- Unreliable information sources - Mixes expert/non-expert content
Standard RAG	Corpus-based retrieval	- Scattered domain information - Unstructured expert knowledge	- User prompt - Retrieved context	- Poor information retrieval to address complex reasoning - Limited context understanding
Proposed Method	Knowledge-based retrieval	- Structured domain information - Organized expert knowledge	- User prompt - Retrieved context	- High initial costs - Requires ongoing maintenance

Previous approaches to contract analysis face significant limitations:

- Base LLMs rely solely on basic user prompts, resulting in hallucination issues and high costs for domain-specific fine-tuning.
- Search-Augmented LLMs incorporate web search but suffer from unreliable information sources and difficulties

distinguishing between expert and non-expert content.

- Standard RAG systems, while more advanced, are ineffective in handling complex contractual reasoning due to poor retrieval performance with scattered domain information and unstructured knowledge.

The proposed method addresses these limitations through knowledge-based retrieval that targets specific domain information and utilizes organized expert knowledge. By integrating a structured knowledge base into the RAG pipeline, the system enhances the precision and reliability of contract analysis. The knowledge base, structured around SCFs, provides a clear framework for systematic analysis of various types of construction contracts through standardized approaches.

While the proposed approach requires significant initial investment and ongoing maintenance, these challenges are mitigated through phased implementation, industry collaboration, and systematic update processes aligned with contract standard revisions. This integrated approach ensures high accuracy in contract analysis and delivers transparent and traceable results, effectively addressing the complexities of construction contract review processes.

By maintaining the existing thesis structure and enhancing the clarity and coherence of the content, this revised section emphasizes the innovative aspects of the CORA-RAG system. It highlights how the integration of structured domain knowledge, advanced retrieval mechanisms, and reliable generation processes overcomes the limitations of existing methods, thereby improving the practical applicability of automated contract review tools in construction projects.

3.3 Knowledge Base Component

This section addresses the first requirement: the systematic integration of expert knowledge to support accurate interpretation and analysis of contract clauses. The knowledge base incorporates expert knowledge organized around the SCFs, mapping each clause to relevant expert analyses, legal information, and practical considerations. Developed based on FIDIC contract guidebooks by Robinson (2013), Seppälä (2023), and Grose (2016), the knowledge base serves as a comprehensive resource providing detailed insights for each contract clause.

For each contractual provision, the system maintains comprehensive documentation, including legal precedents and historical dispute records,

tailored to the complexity of the information. The knowledge selection focuses on construction contract-specific information essential for clause interpretation, understanding legal implications, and addressing practical implementation issues. This structured approach enables comprehensive coverage of contract scenarios, ensuring that the system can interpret complex contractual relationships with precision.

Figure 3 illustrates the process of integrating knowledge into the SCF classification within the knowledge base. The workflow encompasses data collection from both SCFs and related guidebooks, highlighting the extraction of relevant information for each clause. Text preprocessing and information processing stages prepare the data by cleaning and standardizing it. The aggregation stage consolidates this information, aligning it with the knowledge base structure through the SCF classification. This classification ensures that each clause is precisely categorized and linked, facilitating comprehensive contextual analysis that considers the context of each clause. By employing the SCF classification, the system systematically organizes contract clauses, ensuring that each clause is accurately mapped to its corresponding knowledge.

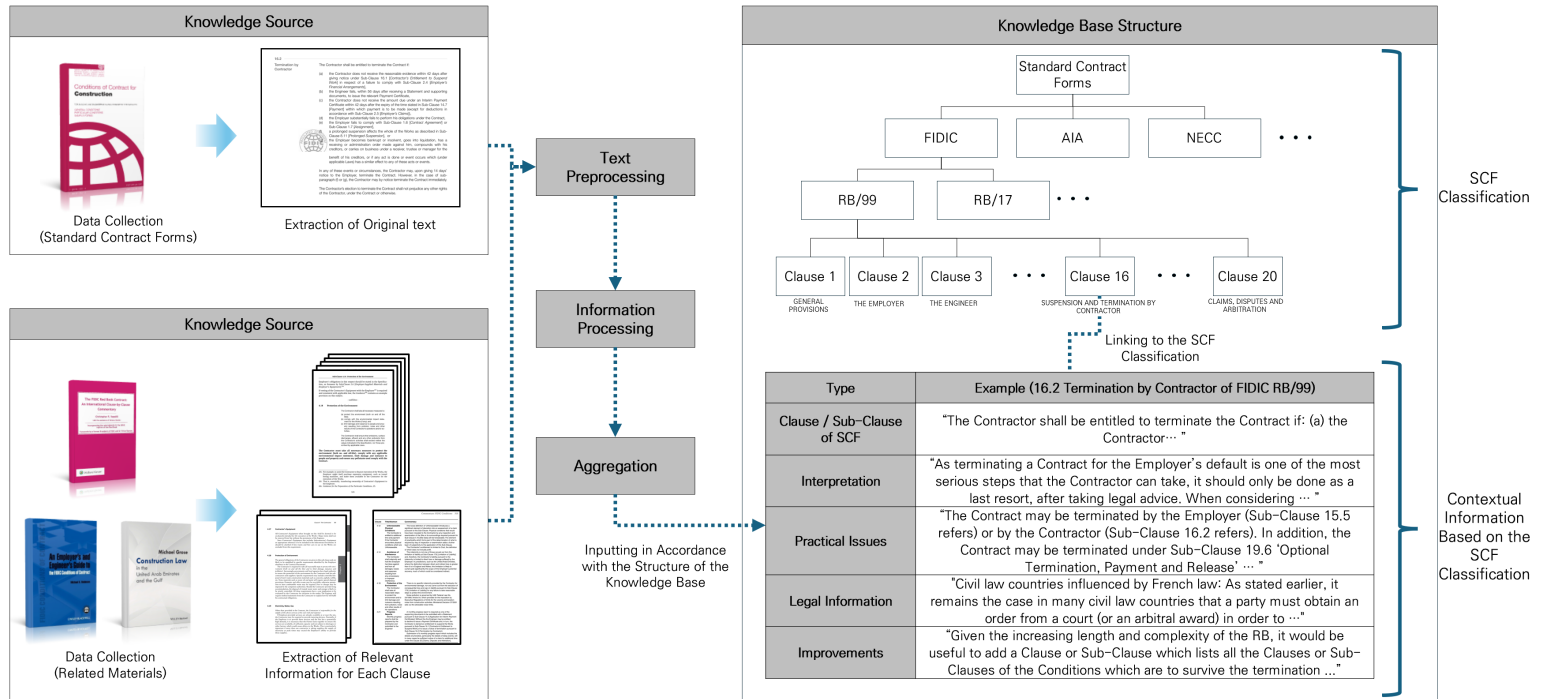


Figure 3. Structured Integration Process of Knowledge into the SCF Classification within the Knowledge Base

Table 4 presents the composition of the knowledge base, integrating each SCF clause or sub-clause with specific types of information. The knowledge is structured around the FIDIC Red Book 1999, providing a detailed framework for each contract clause.

Table 4. Composition the knowledge base

Type	Description	No. of corpus	Average of text length
Clause / Sub-Clause of SCF	Clause text from Standard Contract Forms, providing the base reference for analysis and comparison	162	168
Expert analysis	Detailed explanations of contractual requirements and obligations	162	1488
Legal Information	Case law, legal requirements, and jurisdictional considerations	55	568
Practical Considerations	Common challenges and historical problems	162	138

Each sub-clause of the FIDIC Red Book is systematically documented. The original contract text averages 168 words, detailed expert analysis averages 1,488 words, and practical considerations average 138 words. Legal information is provided through 55 legal documents, each averaging 568 words.

The system transforms this information into machine-friendly formats through a knowledge representation framework that links each clause to its corresponding structured information. This integration enables context-aware contract analysis that considers both practical implications and the literal contract language. To maintain the quality of the knowledge base, the framework includes processes for incorporating updates from FIDIC contract revisions and mechanisms for integrating project feedback and lessons learned.

3.4 Retrieval Component

This section covers the second requirement, inclusion of precise retrieval mechanisms. Combining classification-based filtering with hybrid search methods, this component accurately and efficiently retrieves relevant information from complex contract documents.

The retrieval mechanism operates as a two-stage process. First, clause classification based on SCF classification narrows the search scope within the Knowledge Base, ensuring that only pertinent clauses are considered, reducing noise and improving efficiency. Second, a hybrid search combines dense vector embeddings and sparse keyword-based methods to

precisely identify relevant information. Dense retrieval implements FAISS with the text-embedding-3-small model for semantic similarity searches, mapping textual data into high-dimensional vector space to capture contextual meaning. Sparse retrieval employs BM25—a well-established TF-IDF-based function that ranks documents according to query terms.

This two-stage retrieval mechanism effectively targets specific domain knowledge, enhancing the accuracy and reliability of contract analysis. By processing input contracts through this combined classification and hybrid retrieval method—integrating both dense and sparse retrieval techniques with relationship-aware ranking that considers interconnections among contract elements—the system effectively identifies relevant clauses and associated expert knowledge.

Furthermore, using SCF classification in the initial filtering reduces the search space and aligns the retrieval process with standardized contract structures. This alignment is crucial for maintaining consistency and ensuring that the retrieved information is contextually relevant. By structuring the retrieval mechanism in this way, the system efficiently handles the complexity and length of construction contracts, providing accurate and reliable analytical results.

3.5 Generation Component

This section addresses the third requirement: generating accurate and traceable answers by maintaining connections to reference sources. The Generation Component is responsible for producing contract analyses based on the information prepared by the Retrieval Component and the structured knowledge base.

After the Retrieval Component identifies relevant clauses and associated expert knowledge, the Generation Component performs a differential analysis between the input contract clauses and the corresponding clauses from the SCFs. By comparing the input contract to the SCFs, it identifies discrepancies such as modifications, additions, or omissions. Utilizing contextual information from the knowledge base—including expert analyses, legal information, and practical considerations—the Generation Component assesses the implications of these discrepancies.

To ensure transparency and trustworthiness, each conclusion and recommendation is explicitly linked to specific sections of the knowledge base and SCF clauses. This explicit linkage allows users to easily trace the generated analyses back to their underlying sources, enhancing the reliability and credibility of the system's outputs.

Figure 4 illustrates the sequence diagram of the proposed method, detailing the interactions between the system components during the contract analysis process. The process begins when the user submits an input contract to the system. The System Controller receives the contract and orchestrates the analysis by coordinating with the Retrieval Component. The Retrieval Component classifies the input contract clauses based on the SCF classification to narrow down relevant sections. It then performs a hybrid retrieval using both dense vector embeddings and sparse keyword-based searches to fetch pertinent information from the knowledge base.

Subsequently, the Generation Component compares the input contract clauses with the corresponding SCF clauses to identify differences. It integrates contextual information such as expert analyses, legal considerations, and practical issues to evaluate the implications of these discrepancies. The system then generates detailed analyses and recommendations, explicitly referencing the relevant sources from the knowledge base and SCFs.

Finally, the analyzed output is delivered to the user, providing transparent and traceable insights into the deviations from the SCFs. By maintaining clear connections to reference sources throughout the analysis, the Generation Component produces outputs that are not only accurate but

also verifiable and actionable in professional settings. This approach enhances trust in the automated analysis and supports users in making informed decisions based on reliable and well-documented information.

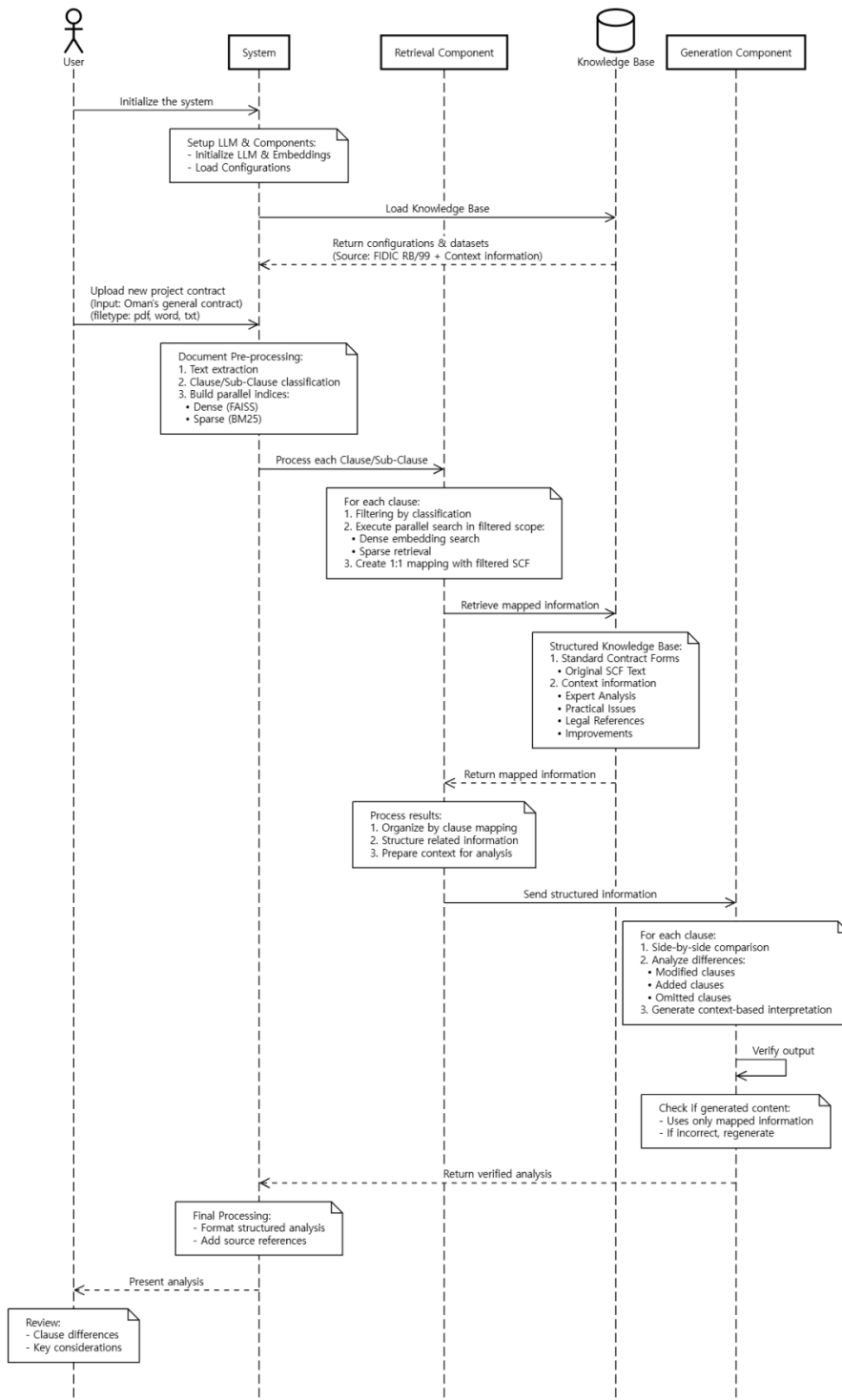


Figure 4. Sequence diagram of proposed method

3.6 Experimental Validation for Retrieval

3.6.1 Evaluation Method and Objectives

In this study, the Retrieval Augmented Generation Assessment System (RAGAS) comprehensively assesses the performance of four approaches: Base LLM, Search-Augmented LLM, Basic RAG, and the proposed method that incorporates structured information. This assessment demonstrates how effectively structured domain knowledge integrates into the RAG for construction contract analysis.

Test cases were designed to evaluate the system's capability to analyze contract clauses by considering multiple aspects, including expert analyses, legal information, and practical considerations. Each test case provides detailed ground truth answers that encompass all these aspects, ensuring that the system's responses can be thoroughly evaluated for completeness and accuracy.

3.6.2 Dataset

The test dataset comprises two widely recognized construction contract documents: the FIDIC Red Book (1999 version) and Oman's General Contract (2019 version), as detailed in Table 5. The FIDIC Red Book,

comprising 20 main clauses and 162 sub-clauses, was chosen because it is widely used and contains extensive information. Oman's General Contract, featuring 22 main clauses and 167 sub-clauses, was selected to assess the system's adaptability to real-world contract. After excluding definition sections, both documents yielded a comprehensive corpus of 2,071 text segments for analysis.

Table 5. Overview of Test Dataset

Document Type	No. of Clause / Sub-Clause	No. of corpus	No. of words	No. of tokens
FIDIC Red Book	20 / 162*	1021	29280	35574
Oman's general contract (test)	22 / 167*	1050	33999	40809

* Excludes definitions section

3.6.3 System Implementation and Comparison

The experimental setup involves comparing four approaches, as detailed in Table 6. The Base LLM employs the GPT-4o-mini-2024-07-18 model as a benchmark. The Search-Augmented LLM uses web search using the Perplexity API with llama-3.1-sonar-large-128k model. The Basic RAG and the proposed method implement a hybrid retrieval system combining dense and sparse retrieval techniques using FAISS and BM25.

All approaches, excluding Perplexity, use the same base LLM (GPT-4o-mini-2024-07-18) to ensure consistent comparison. This consistent baseline allows for evaluation of each approach's effectiveness in contract analysis tasks, demonstrating the advantages of integrating structured domain knowledge into the RAG pipeline.

Table 6. Experimental Setup

Approach	Base Model	Retrieval Configuration
Base LLM	GPT-4o-mini-2024-07-18	N/A
Search-Augmented LLM	Llama-3.1-sonar-large-128k	Web Search (Perplexity API) Top-k: 5
Basic RAG	GPT-4o-mini-2024-07-18	Dense: FAISS, text-embedding-3-small Sparse: BM25, k1=1.5, b=0.75 Hybrid weights (Dense:0.5, Sparse:0.5) Top-k: 5
Proposed Method	GPT-4o-mini-2024-07-18	Dense: FAISS, text-embedding-3-small Sparse: BM25, k1=1.5, b=0.75 Hybrid weights (Dense:0.5, Sparse:0.5) Top-k: 5

3.6.4 Evaluation Metrics

The evaluation method utilizes RAGAS metrics, enabling assessment without reliance on ground truth human annotations (Es et al., 2023). These metrics evaluate both retrieval and generation capabilities and are categorized into retrieval performance, response quality, and overall system effectiveness, as detailed in Table 7.

Context Precision measures how relevant the retrieved information is to the query, ensuring the system focuses on pertinent contract clauses without unnecessary content. Context Recall evaluates the completeness of the retrieved information, confirming that all critical clauses needed to answer the query are included.

Answer Relevancy assesses how directly the system's responses address the user's specific queries, ensuring practical usefulness for contract reviewers. Faithfulness measures how accurately the responses reflect information from the SCFs and expert knowledge base, maintaining the integrity of contract interpretation.

Answer Correctness combines accuracy and quality to provide an overall measure of the system's practical utility, determining whether the answers are correct and useful for professional contract review. These metrics

assess the system's ability to retrieve relevant contract information and generate accurate, reliable analyses, ensuring that the automated review process meets professional standards.

Table 7. Evaluation metrics and description

Evaluation level	Metric Type	Description	Input Variables	Description
Retrieval	Context, Precision	Relevance of retrieved information	Ground Truth, Context	Measures the proportion of relevant context within the retrieved information
	Context Recall	Completeness of retrieved information	Ground Truth, Context	Assesses the proportion of all relevant context successfully retrieved
Generation	Answer Relevancy	How directly the response addresses the query	Answer, Question	Evaluates alignment between the answer and the question for relevance
	Faithfulness	Consistency with retrieved context	Answer, Context	Checks that the answer accurately reflects the retrieved context
Overall	Answer Correctness	Combined accuracy and usefulness of the answer	Answer, Ground Truth	Determines overall answer accuracy and utility against the ground truth

3.6.5 Results and Analysis

The experimental results reveal significant performance variations across different model approaches, as Table 8 illustrates. Each approaches shows distinct characteristics in both information retrieval and answer generation capabilities.

(1) Information Retrieval Performance

The Base LLM does not provide context retrieval metrics because it operates without accessing external documents. The Search-Augmented LLM, despite having access to external information, shows moderate retrieval capabilities (Context Precision: 0.721, Recall: 0.558) but with high variability (± 0.297 and ± 0.392 respectively), indicating stability issues in retrieval performance.

The Basic RAG implementation utilizes a hybrid retrieval approach combining dense and sparse retrieval methods (Context Precision: 0.587 ± 0.115 , Recall: 0.635 ± 0.363). While this approach enables both semantic understanding and exact matching of contract terms, its Context Precision is lower than the Search-Augmented LLM, though with improved stability as shown by lower standard deviations.

The proposed method achieves high Context Recall (0.969 ± 0.132)

while exhibiting similar Precision (0.625 ± 0.089) compared to the other methods. This superior performance, particularly in stability as indicated by the lowest standard deviations among all approaches, stems from the structured organization of domain knowledge, enabling more comprehensive and reliable retrieval of relevant contract information.

(2) Answer Generation Quality

The Base LLM exhibits strong Answer Relevancy (0.818), reflecting its robust language understanding capabilities developed through pre-training. The Search-Augmented LLM shows improved Answer Relevancy (0.884) and Faithfulness (0.770), indicating effective integration of retrieved information in answer generation. The Basic RAG shows moderate Faithfulness (0.602), indicating integration of retrieved information in answer generation, though not surpassing the Search-Augmented LLM's performance (0.770)

The proposed method significantly enhances both metrics (Faithfulness: 0.893 ± 0.198 , Answer Relevancy: 0.903 ± 0.150), demonstrating effective utilization of structured knowledge in generating accurate and relevant responses. Additionally, the comparatively low standard deviations (± 0.198 for Faithfulness and ± 0.150 for Answer Relevancy) indicate consistent response quality across various queries,

highlighting the reliability of the proposed method.

Furthermore, the proposed method achieves the highest Answer Correctness of 0.750 ± 0.197 , indicating its superior ability to provide accurate and correct responses essential for construction contract review. The integration of structured information in the knowledge base enhances the quality and organization of the content used for answer generation. This leads to significant improvements in both Faithfulness (from 0.602 to 0.893) and consistency of performance, as shown by lower standard deviations across metrics.

Table 8. Performance result of different LLM approaches

	Evaluation Metrics (Mean $\pm$ SD)					
	Context Metrics		Answer Metrics		Overall Performance	
Model Type	Context Precision	Context Recall	Answer Relevancy	Faithfulness	Answer Correctness	Average Score
Base LLM (GPT-4o-mini)	N/A	N/A	0.818 $\pm$ 0.312	N/A	0.687 $\pm$ 0.240	0.752 $\pm$ 0.200
Search-Augmented LLM (Perplexity API, llama-3.1-sonar-large)	0.721 $\pm$ 0.297	0.558 $\pm$ 0.392	0.884 $\pm$ 0.093	0.770 $\pm$ 0.252	0.616 $\pm$ 0.205	0.710 $\pm$ 0.134
Standard RAG (GPT-4o-mini, Dense+Sparse Retrieval)	0.587 $\pm$ 0.115	0.635 $\pm$ 0.363	0.895 $\pm$ 0.148	0.602 $\pm$ 0.249	0.697 $\pm$ 0.213	0.683 $\pm$ 0.119
Proposed Method (GPT-4o-mini, Dense+Sparse Retrieval, Structured information)	0.625 $\pm$ 0.089	0.969 $\pm$ 0.132	0.903 $\pm$ 0.150	0.893 $\pm$ 0.198	0.750 $\pm$ 0.197	0.828 $\pm$ 0.087

3.7 Interim Discussion

3.7.1 Performance Analysis

By fulfilling the three requirements outlined in Section 3.1, the Structured Knowledge-Integrated RAG Pipeline effectively addresses significant challenges in automated contract analysis for construction contracts. Firstly, the systematic integration of expert knowledge through a well-organized knowledge structure enables the system to handle specialized technical and legal tasks with greater accuracy. By incorporating domain-specific expertise into a knowledge base, the system mitigates the limitations of LLMs in dealing with specialized knowledge not widely represented in their training data, ensuring precise interpretation of contract clauses.

Secondly, the incorporation of precise retrieval mechanisms addresses challenges in handling large volumes of unstructured information and maintaining consistent interpretations. By combining classification-based filtering with hybrid dense-sparse retrieval techniques, the system mitigates limitations of basic RAG approaches, improving the retrieval of relevant clauses and contextual information.

Thirdly, maintaining connections to reference sources enhances traceability, mitigating the probabilistic nature of LLM outputs and ensuring

the reliability. Through reference-based generation, the system allows users to trace conclusions back to original sources, strengthening trustworthiness where accuracy is paramount.

3.7.2 Enhancements via Structured Information Integration

The proposed method achieved Context Recall of 0.969 and Faithfulness of 0.893, compared to Basic RAG's 0.602. This demonstrates improved reference maintenance and reduced hallucinations.

Figure 5 illustrates the comparative analysis of responses to "What measures must the Contractor take to protect the environment?" across different models.

The Base LLM offered only general environmental protection practices without addressing specific contractual requirements. It missed critical obligations like taking "all necessary measures," limiting pollution and noise damage, controlling emissions, and complying with the environmental impact statement.

Both the Search-Augmented LLM and Basic RAG included key contractual obligations but diluted their responses with irrelevant information about local regulations, monitoring systems, and legal consequences. This

additional content reduced the precision required in contractual contexts. Moreover, both models exhibited variability in performance metrics. The Search-Augmented LLM demonstrated even more pronounced variability, with a Context Precision standard deviation of ± 0.297 and Context Recall standard deviation of ± 0.392 . This high variability undermines the reliability essential for tasks that require consistent precision, such as contract analysis.

The proposed method often generated responses closely matching the contractual content. It specified that the Contractor must take all necessary measures to protect the environment both on and off site, limit damage from pollution and noise, ensure emissions don't exceed specified values, and comply with environmental impact statements. This demonstrates the effectiveness of integrating structured information with advanced retrieval techniques for contractual content, enhancing consistency and maintaining high recall.

Ground Truth	The Contractor must take all necessary measures to protect the environment, limit damage and nuisance from pollution and noise, ensure emissions and discharges do not exceed specified values, and comply with the environmental impact statement for the Works.
Question	"What measures must the Contractor take to protect the environment?"

	Retrieved Contexts	Generated Answer
Base LLM	N/A	The Contractor must implement measures such as minimizing waste, using environmentally friendly materials, preventing contamination of soil and water, protecting wildlife habitats, and adhering to local environmental regulations and guidelines to protect the environment.
Search-Augmented LLM	[1]: "This Sub-Clause places a significant responsibility on the Contractor to ensure environmental compliance. It emphasizes the importance of adhering to both local environmental laws ..." [2]: "Practically, the Contractor needs to conduct thorough environmental assessments and implement appropriate measures to mitigate any potential environmental harm. ..." ...	The Contractor must protect the environment through these essential measures: <u>The Contractor is required to protect both on-site and off-site environments while implementing necessary measures to minimize pollution, noise, and other operational impacts.</u> They must ensure all emissions, surface discharges, effluent, and pollutants remain within specified limits. Additionally, they <u>must comply with the project's environmental impact statement and take steps to limit damage and nuisance to people and property.</u> The implementation of these measures requires careful attention to local regulations and proper measurement of emissions and pollutants. This responsibility carries legal implications including potential liability for environmental damage and fines for non-compliance.
Basic RAG	[1]: "The Contractor shall take all necessary measures to protect the environment (both on and off the Site), comply with the environmental impact statement for the Works (if any), ..." [2]: "The Sub-Clause emphasizes the Contractor's responsibility for environmental protection and compliance with regulations and specified standards. It reflects a growing recognition of ..." ...	The Contractor must protect the environment through several key measures: <u>The Contractor is required to protect both on-site and off-site environments while implementing necessary measures to minimize environmental impact.</u> They must <u>strictly comply with the project's environmental impact statement and control pollution, noise, and other operational impacts that could affect people and property.</u> A crucial responsibility is <u>maintaining emissions, surface discharges, effluent, and pollutants within specified limits.</u> This includes <u>adhering to local environmental regulations</u> and implementing proper monitoring systems.The Contractor faces potential legal consequences for environmental violations, including fines and liability for damages, making strict compliance essential.
Proposed Model	[1]: "Expert Analysis: (1) Interpretation of Sub-Clause While this Sub-Clause is drafted very broadly (providing that the Contractor shall take all necessary measures to [...] protect ..." ... [4]: "Legal Information: An EIA is now required by the domestic law of many countries. In addition, the Rio Declaration resulting from the United Nations Conference on Environment and Development in Rio de Janeiro, 1992, confirmed that EIAs ..." [5]: "Practical Considerations: The first sentence of this Sub-Clause should ..."	The Contractor must take the following environmental protection measures: <u>Take all necessary measures to protect the environment both on and off the Site, limiting damage and nuisance to people and property from pollution, noise, and other operational impacts.</u> <u>Ensure that emissions, surface discharges, and effluent from construction activities do not exceed specification values or applicable legal limits.</u> <u>Comply with any environmental impact statements for the Works.</u> The obligation covers environmental protection during contract performance, though the requirement has been strengthened from "reasonable steps" to "necessary measures". The Contractor must also <u>follow domestic environmental laws and international standards, particularly regarding Environmental Impact Assessments.</u>

Figure 5. Comparison of Retrieved Contexts and Generated Answers by Different Models

3.7.3 Limitations

Despite the strong performance, certain limitations highlight areas for improvement. First, although the system achieves a high Context Recall (0.969), the relatively lower Context Precision of 0.625 indicates that the retrieval mechanism occasionally includes less relevant information. Additionally, the Answer Correctness metric of 0.750 suggests potential for improvement in generating accurate responses.

Second, although the RAGAS evaluation offers valuable quantitative insights into the system's retrieval and answer generation performance, it does not fully capture the nuanced judgments of human experts, highlighting the necessity for manual expert evaluations to ensure comprehensive assessment of analysis quality.

Third, the system's effectiveness depends on the comprehensiveness of its knowledge base for SCFs, which may limit its applicability to contracts outside of this scope. This reliance on a specific knowledge base means the system may not be applicable to tailored contracts that are customized or significantly modified.

Future research will focus on expanding the knowledge base to cover a wider range of contract types, including customized and non-standard

contracts. Enhancing the knowledge base and advancing retrieval algorithms are anticipated to bridge the remaining gaps. Moreover, integrating human expert manual assessments will be crucial for advancing the system's reliability and effectiveness in contract analysis.

4. Generate-Focused Framework

4.1 Multi-agent LLM Framework

4.1.1 Motivation for Refinement and Generation

The initial CORA-RAG system automated parts of the construction contract review process, yet challenges persist in the generation phase due to limitations of current LLMs. These models often struggle with logical reasoning and understanding cause-and-effect relationships within contract terms. They may identify changes in specific clauses but fail to comprehend how these modifications impact interconnected provisions.

To overcome these limitations, a multi-agent LLM framework is introduced. This approach employs multiple LLM agents assigned to specific tasks or domains—such as differential analysis, or risk assessment—enhancing focus and expertise without altering the underlying models. Instead of fine-tuning, strategic prompt engineering and task-specific configurations are utilized to tailor each agent's function.

By assigning specific roles to LLM agents and enabling collaboration, the system can better handle complex contractual contexts. Agents share information and build upon each other's outputs, distributing the workload and improving reasoning. Key aspects of this approach include:

- Task-specific LLM Agents through Prompt Engineering: Directing each agent to focus on a specific task using tailored prompts and instructions.
- Collaborative Workflow: Coordinating agents through a central system controller that manages data exchange and ensures a coherent analysis process.
- Integration of Domain Expertise: Incorporating expert knowledge into the agents' configurations to ensure accurate and in-depth analysis.

Figure 6 illustrates how this system mirrors traditional expert review practices. It integrates LLM agents at key processing stages corresponding to expert tasks like clause identification, contextual analysis, and impact assessment. The left side of the figure represents the traditional sequential review process, forming the foundation of the automated framework. The right side shows the AI-based implementation with task-specific LLM agents.

The multi-agent approach improves analysis quality by enabling a better understanding of complex contractual contexts. This ensures that automation maintains the review's depth and quality, leading to accurate and reliable contract analyses aligned with professional standards.

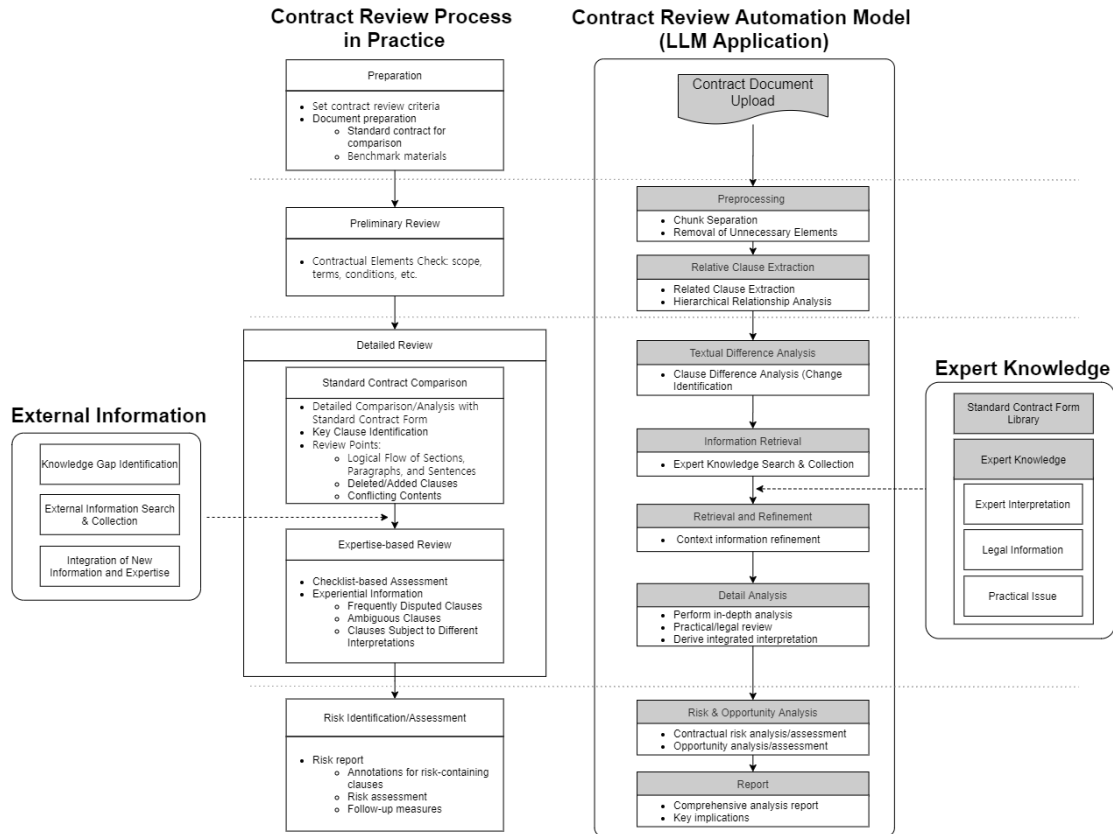


Figure 6. Simulating Contract Review Process in Practice

4.1.2 System Requirements for Multi-Agent LLM Framework

The system employs a coordinated network of LLM agents operating in a structured, sequential workflow to perform comprehensive contract analysis. As illustrated in Figure 7, ensuring effective communication and data flow between agents while maintaining overall process integrity. Validation mechanisms at each stage ensure outputs are reliable and accurate before proceeding to the next phase.

1. Agent Deployment and Coordination: The system must support the deployment and coordination of various LLM agents, each assigned to specific tasks such as textual difference analysis, semantic analysis, context analysis, detailed examination, summary generation, risk-opportunity assessment, and results integration. Agents operate sequentially and collaboratively, with each agent's output serving as input for the next. The central system controller facilitates communication among agents, monitors progress, and handles exceptions to ensure workflow continuity.

2. Validation Framework: The system incorporates robust validation mechanisms to verify outputs at each analysis stage. This includes automated checks performed by LLM agents and interfaces for domain experts to review and confirm analyses. The absence of robust validation processes involving

domain experts resulted in outputs that, although helpful for basic analysis, fell short in handling complex legal interpretations. Enhancing validation by incorporating interfaces for domain experts to review and confirm analyses is necessary. Such validation will ensure that each agent's contribution aligns with professional standards of accuracy and relevance, thereby improving the system's overall reliability before proceeding to subsequent analysis stages.

3. Traceability: The system maintains traceability and transparency in the reasoning process, allowing users to understand how conclusions are reached at each stage. This enhances trust by making the analysis explainable, enabling stakeholders to follow the reasoning behind insights and decisions. Detailed logs and documentation capture each agent's inputs, outputs, and processing steps, facilitating traceability.

These requirements establish a foundation for effective collaboration among LLM agents, ensuring a smooth and coherent workflow while maintaining reliability and transparency throughout the contract analysis process.

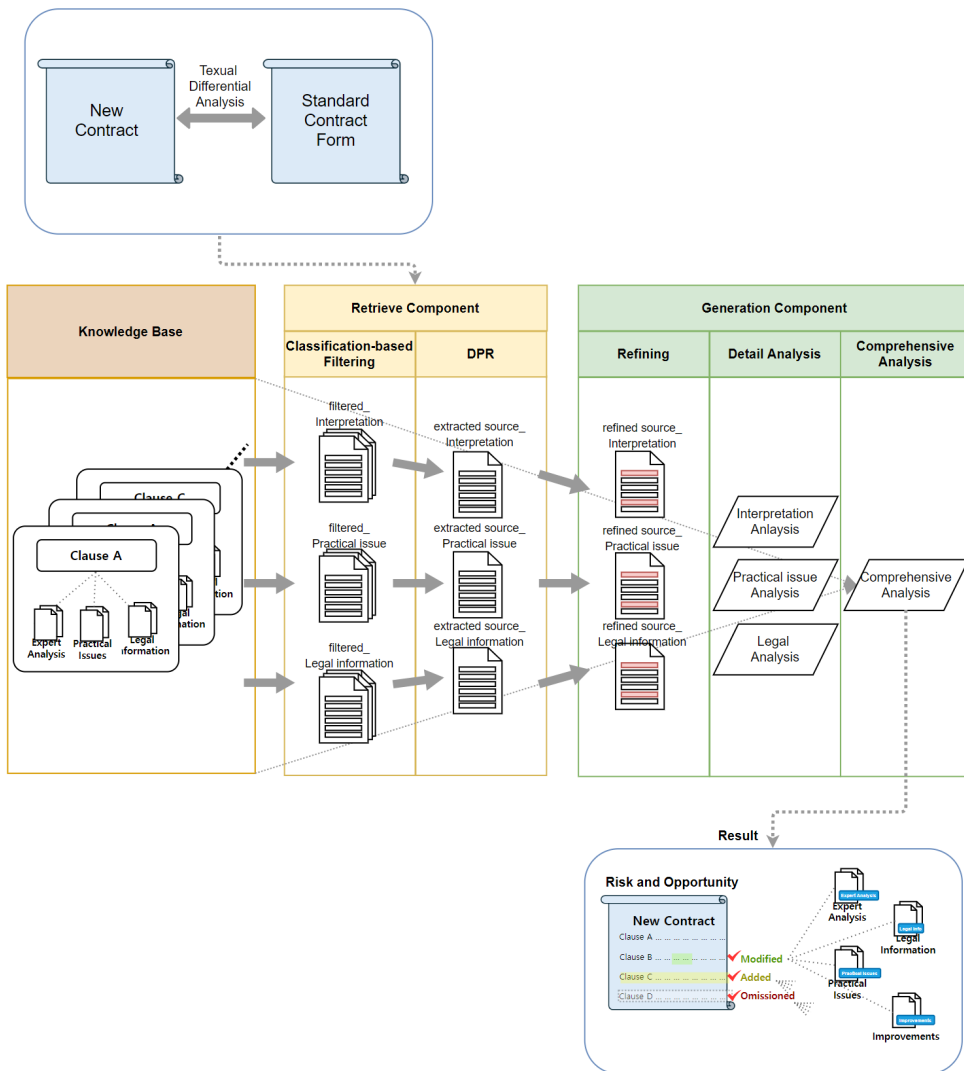


Figure 7. Multi-stage Analysis Process of CORA-RAG System

4.2 Generation System Architecture

4.2.1 Context Refinement Pipeline

The Context Refinement Pipeline is a critical component of the CORA-RAG system, integrating specialized technical and legal knowledge essential for construction contract reviews. As depicted in Figure 7, this pipeline operates through a network of LLM agents performing specific tasks, with certain processes operating in parallel to enhance efficiency and depth.

The process begins with the Textual Difference Analysis Agent, which identifies significant variations between the new contract clauses and the standard clauses from the FIDIC RED99. This agent processes inputs from both the new contract and the standard contract library to generate a list of detected changes with initial annotations. These changes highlight modifications in contractual language and provisions that require further examination.

Following the identification of differences, the pipeline advances into two parallel refinement processes: Semantic Analysis and Context Analysis.

- **Semantic Analysis Agent:** This agent focuses on examining the language modifications identified by the Textual Difference Analysis Agent. It assesses how these changes impact contractual obligations and rights, providing

semantic insights into the contractual modifications. This analysis deepens the understanding of the altered meanings and intentions behind the revised clauses.

- **Context Analysis Agent:** Utilizing expert knowledge bases—including expert analysis, legal information and practical issues—this agent evaluates the legal requirements and practical considerations associated with the contractual changes. It offers contextual insights that illuminate the broader implications of the modifications, such as compliance with laws and the feasibility of implementation.

The outputs from the Semantic Analysis Agent and the Context Analysis Agent are then combined by the Combined Refined Context Agent to form a comprehensive refined context. This agent synthesizes the semantic understanding of the language changes with the practical and legal considerations, providing a robust foundation for the subsequent stages of analysis. Validation mechanisms are employed at each stage to ensure the accuracy and reliability of the information before it progresses further in the pipeline.

By operating the Semantic Analysis and Context Analysis in parallel, the system enhances both efficiency and depth in processing contractual modifications. This approach allows for a thorough understanding of complex contractual changes, mirroring professional review practices. The refined context produced at the end of this pipeline equips the system to proceed with detailed examination and risk-opportunity assessments, ultimately supporting stakeholders in making informed decisions regarding contract management.

Table 9. Context Refinement Pipeline Agents

Agent	Input	Function	Output
Differences Analysis Agent	Input contract, SCF	Identifies significant variations between new and standard clauses	List of detected changes with initial annotations
Semantic Analysis Agent	Detected changes from Differences Analysis	Examines language modifications and assesses implications on obligations	Semantic insights into contractual modifications
Context Analysis Agent	Detected changes from Textual Difference Analysis, Expert knowledge bases	Integrates expert analysis, legal implications and practical considerations	Contextual insights related with SCF's conditions
Combined Refined Context Agent	Outputs from Semantic and Context Analyses	Combines insights to form a comprehensive refined context	Comprehensive refined context for further analysis

4.2.2 Multi-stage Generation Process

After context refinement, the system proceeds to the Multi-Stage Generation Process, where LLM agents build upon the refined context to perform deeper analyses and generate outputs. As detailed in Table 10, these agents collaborate sequentially to provide users with a consolidated report that includes technical, legal, practical, and strategic recommendations tailored to specific contract scenarios.

The Detailed Examination Agent utilizes the accumulated refined context to perform a thorough examination of the contractual provisions. It produces detailed analysis results, highlighting how the changes materially affect the contractor's obligations and what adjustments may be necessary to meet the new standards.

Following this, the Summary Generation Agent synthesizes the detailed examination results into concise summaries. This agent distills key findings, allowing stakeholders to grasp the essential points without delving into extensive details. The summaries provide quick overviews that facilitate rapid understanding of the significant enhancements in obligations and their potential impact on responsibilities.

Building upon the previous analyses, the Risk-Opportunity

Assessment Agent evaluates the contractual changes to identify key considerations. This agent produces both risk assessments and opportunity assessments:

- Risk Assessment: Identifies and articulates legal, operational, and financial risks associated with the contractual modifications. For example, it notes increased legal exposure for the contractor due to potential remediation costs stemming from stricter environmental requirements.
- Opportunity Assessment: Highlights beneficial aspects and strategic advantages that the contractual changes may bring. It identifies opportunities such as enhanced market positioning as a responsible contractor focused on sustainability, which could lead to an improved reputation and competitive edge in sustainability-focused markets.

Finally, the Results Integration Agent consolidates all analysis outputs. It gathers results from the previous agents, including detailed examinations, summaries, and risk-opportunity assessments, and performs comprehensive analysis integration and report generation. This agent ensures

that all insights are coherently presented, providing a unified perspective on the contractual changes. The output is an integrated analysis report with key recommendations, offering actionable advice that aligns technical, legal, practical, and strategic considerations to support informed decision-making.

Table 10. Multi-stage Generation Process Agents

Agent	Input	Function	Output
Detailed Examination Agent	Accumulated refined context	Performs thorough examination and provides insights into contractual provisions	Detailed analysis results
Summary Generation Agent	Detailed analysis results	Synthesizes detailed examination results into concise summaries	Summary results: quick overviews distilling key findings
Risk-Opportunity Assessment Agent	Outputs from previous analyses	Evaluates contractual changes to identify risks and opportunities	Risk and opportunity assessments: identification of risks and strategic advantages
Results Integration Agent	Outputs from previous analyses	Integrates all analysis outputs and generates a comprehensive report	Integrated analysis report with key recommendations

4.2.3 System Implementation

The system integrates the Context Refinement Pipeline and the Multi-Stage Generation Process into an efficient, cohesive workflow, as

depicted in Figure 8. This workflow mirrors traditional expert analysis practices while leveraging the capabilities of LLM agents. Agents are assigned specific tasks within a sequential and collaborative process, each building upon the previous agent's output to ensure depth and accuracy in contract analysis.

The implementation begins with the Textual Difference Analysis Agent identifying significant variations between the new contract clauses and the standard clauses. This agent sets the foundation for deeper analysis by pinpointing the exact changes in contractual language that require further examination.

Following this, the Semantic Analysis Agent and the Context Analysis Agent operate in parallel to refine the context:

- The Semantic Analysis Agent examines the language modifications identified, assessing how these changes impact contractual obligations and rights. It provides semantic insights into the contractual modifications, enhancing the understanding of altered meanings and intentions behind the revised clauses.
- The Context Analysis Agent integrates domain knowledge,

including relevant laws, regulations, and industry practices, to evaluate the legal requirements and practical considerations associated with the contractual changes. It offers contextual insights that illuminate the broader implications of the modifications, such as compliance issues or operational challenges.

Their outputs are then combined by the Combined Refined Context Agent to form a comprehensive refined context. This agent synthesizes the semantic understanding of the language changes with the practical and legal considerations, providing a robust foundation for the subsequent stages of analysis. Validation mechanisms at each stage ensure the accuracy and reliability of the information before it progresses further in the workflow.

Next, the Detailed Examination Agent utilizes the refined context to perform an in-depth analysis of the contractual provisions. It highlights how the changes materially affect obligations and identifies necessary adjustments to meet new standards. This agent delves into the practical impact of the contractual modifications on responsibilities and requirements.

The Summary Generation Agent then synthesizes the detailed examination results into concise summaries. It distills key findings, allowing

stakeholders to quickly grasp essential points without delving into extensive details. This facilitates rapid understanding of significant enhancements in obligations and their potential impact.

Building upon these analyses, the Risk-Opportunity Assessment Agent evaluates the contractual changes to identify potential risks and opportunities:

- It assesses risks by identifying increased legal exposure, potential remediation costs, and other legal, operational, and financial risks associated with the contractual modifications.
- It identifies opportunities by highlighting beneficial aspects and strategic advantages, such as enhanced market positioning or improved reputation resulting from compliance with higher standards.

Finally, the Results Integration Agent consolidates all analysis outputs. It gathers results from the previous agents—including detailed examinations, summaries, and risk-opportunity assessments—and generates an integrated analysis report with key recommendations. This agent ensures that all insights are coherently presented, providing a unified perspective on

the contractual changes and offering actionable advice that aligns technical, legal, practical, and strategic considerations.

The overall process flow is managed by the system controller, which coordinates the sequence of operations, facilitates communication among agents, monitors progress, and handles exceptions to maintain workflow integrity. This structured workflow is illustrated in Figure 8, showing how the agents collaborate from the initial identification of textual differences to the final generation of the comprehensive analysis report.

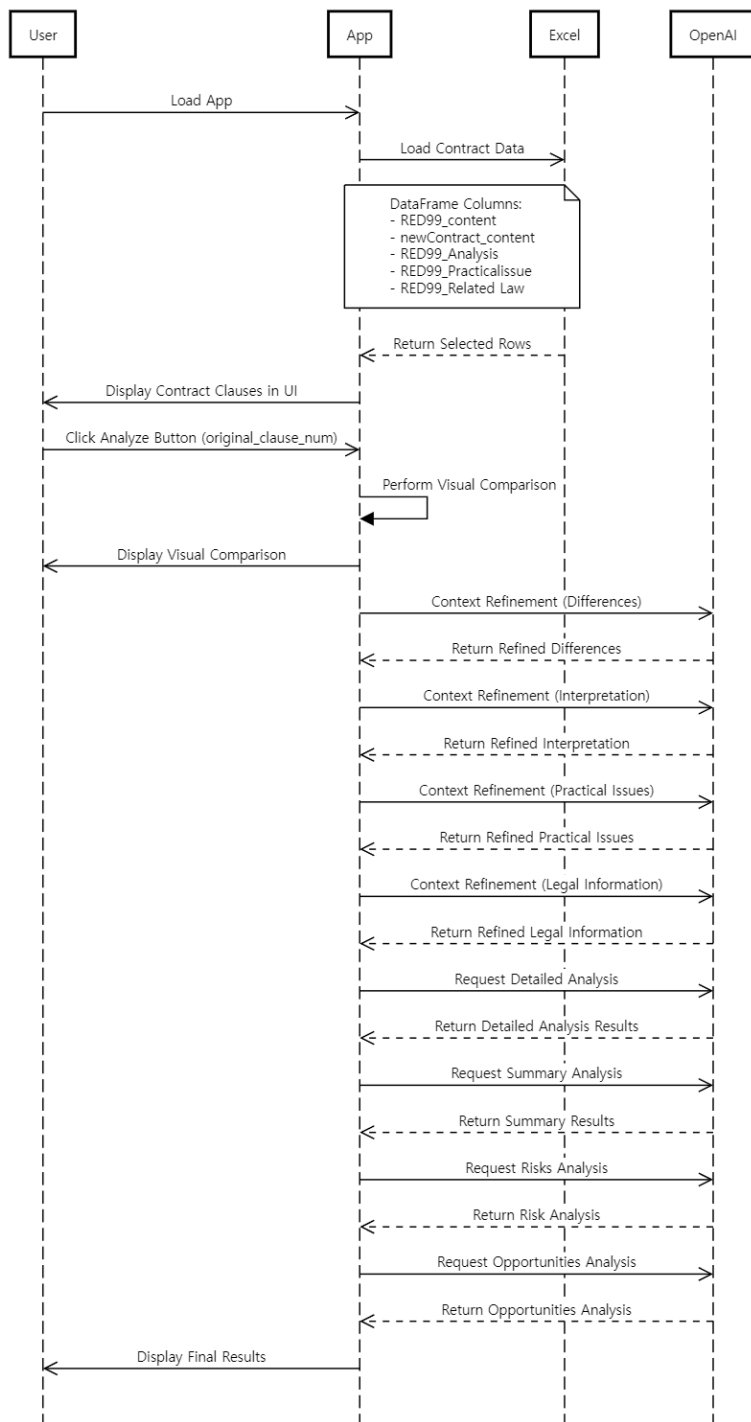


Figure 8. Refinement and Analysis Workflow

4.2.4 User Interface of CORA-RAG System

The CORA-RAG system provides an interface that organizes relevant contract clauses and facilitates efficient navigation of the analysis results. Users can focus on specific clauses and view key differences highlighted through visual comparisons, enabling them to quickly identify and assess modifications. This interface presents the outputs generated at each stage of the analysis process, displaying detailed examinations, summaries, risk assessments, opportunity identifications, and key recommendations produced by the LLM agents.

Figures 9 and 10 illustrate how the multi-stage contract analysis process is presented within the user interface:

- Figure 9 depicts the sequential procedure starting with side-by-side comparisons of contract clauses. It shows how the Textual Difference Analysis Agent highlights modified clauses and how subsequent agents extract relevant data and refine the context. The interface displays these stages, allowing users to follow the progression of the analysis and understand how each agent contributes to the overall process.

- Figure 10 presents the outcomes of higher-level assessments generated by the LLM agents. It displays the integrated analysis report, offering a comprehensive perspective on the contractual changes. This includes detailed analyses, summaries, and evaluations of risks and opportunities, providing stakeholders with a unified overview of the analysis results and the implications of the contractual modifications.



Figure 9. Sequential analysis process in CORA-RAG system

(a) Result of comprehensive analysis

```

{
  "Key_Changes": [
    {
      "type": "added"
      "source_text":
        "each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment."
      "description":
        "This addition clarifies that payments made to the Contractor satisfy the Employer's obligations without any claims from joint venture members regarding payment distribution."
      "impact": {
        "interpretation": {
          "summary_content":
            "The clause limits the rights of JV members to challenge payment distributions, enhancing clarity in financial transactions."
          "reference": {...}
        }
        "practical": {...}
        "legal": {...}
        "significance": "high"
        "criteria":
          "This change significantly impacts financial relationships and potential disputes among JV members."
      }
    }
  ]
}

```

(b) Report

Risk and Opportunity Analysis							
Sub-Clause No.	Title	Text Difference	Type	Description	Importance	Reasoning	Evidence
0 1.14	Joint and Several Liability	[added]each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.[/added]	Risk	Payments made to the Contractor are considered fully satisfied regardless of internal division among JV members, which may lead to disputes among members regarding the distribution of funds received.	Medium	{ "Interpretation": "The addition clarifies payment obligations, limiting the rights of JV members to challenge payment distributions.", "Practical": "This change may lead to disputes among JV members regarding the distribution of funds received, affecting relationships and project dynamics.", "Legal": "Establish clear payment processes and communication among members to avoid disputes over payment distribution." }	{ "Interpretation": "(1) The Contractor as a Joint Venture: This Sub-Clause should be read together with the definitions of a \u2018JV\u2019 in Sub-Clause 1.1.46 and of a \u2018JV Undertaking\u2019 in Sub-Clause 1.1.47.370 As projects become larger and more complex, it is increasingly common for the Contractor to take the form of a joint venture (or \u2018JV\u2019) in Sub-Clause 1.1.47.370. (2) Joint and several liability: Sub-paragraph (a) provides that the JV members shall be \u2018jointly and severally\u2019 liable to the Employer for the performance of the Contractor\u2019s obligations.", "Practical": "This may lead to disputes among JV members regarding the distribution of funds received, affecting relationships and project dynamics.", "Legal": "Establish clear payment processes and communication among members to avoid disputes over payment distribution." }

Figure 10. Analysis Results in CORA-RAG System

4.2.5 Detailed Architecture Implementation and Case Analysis

The CORA-RAG system demonstrates its effectiveness through the analysis of Sub-Clause 4.18 "Protection of the Environment" from the Oman case construction contract. The system processes this analysis through its LLM agents in a structured workflow. The system conducts its analysis through several key stages:

(1) Initial Analysis:

- The process begins with the Textual Difference Analysis Agent identifying significant variations between the standard FIDIC RED99 clauses and the new contract clauses. Notably, the new clause imposes stricter environmental obligations, particularly regarding emissions and discharge controls. By meticulously comparing the texts, the agent highlights these differences, setting the foundation for deeper analysis.

(2) Context Refinement:

- Following the identification of differences, the system undertakes two parallel refinement processes: Semantic Analysis and Context Analysis. In Semantic Analysis, the

system examines the language modifications and their implications on contractual obligations and rights. This examination provides insights into how the changes affect the interpretation of the clause, clarifying nuanced shifts in meaning that could impact the contract's execution.

- Simultaneously, Context Analysis integrates domain knowledge, including legal requirements and practical considerations. The system assesses relevant laws, regulations, and industry practices to understand the broader implications of the contractual changes. This ensures that the modifications are evaluated within the appropriate legal and practical framework, highlighting potential compliance issues or operational challenges.

(3) Detailed Examination:

- With refined information from both analyses, the Detailed Examination Agent produces comprehensive interpretations. For example, it highlights that "The Contractor's responsibilities regarding environmental damage are intensified," emphasizing enhanced accountability and operational requirements. This stage

deepens the understanding of how the changes materially affect the contractor's obligations and what adjustments may be necessary to meet the new standards.

(4) Summary Generation:

- To facilitate quick understanding, the Summary Generation Agent synthesizes the detailed examination results into concise summaries. It distills key findings, allowing stakeholders to grasp the essential points without delving into extensive details. An example output includes an overview highlighting the significant enhancements in environmental obligations and their potential impact on contractor responsibilities. This summary aids decision-makers in quickly assessing the implications of the contractual changes.

(5) Risk-Opportunity Assessment:

- Building upon the detailed examination, the Risk-Opportunity Assessment Agent evaluates the contractual changes to identify key considerations. Regarding risks, it notes increased legal exposure for the contractor due to

potential remediation costs, outlining potential legal and financial risks associated with stricter environmental requirements. In terms of opportunities, it identifies an enhanced market position as a responsible contractor focused on sustainability, highlighting strategic advantages such as improved reputation and competitive edge in sustainability-focused markets. This assessment provides balanced insights into potential challenges and benefits arising from the contractual modifications.

(6) Results Integration:

- In the final stage, the Results Integration Agent consolidates all analysis outputs. It gathers results from the previous agents, including detailed examinations, summaries, and risk-opportunity assessments. By performing comprehensive analysis integration and report generation, the agent ensures that all insights are coherently presented, providing a unified perspective on the contractual changes.
- The output is an integrated analysis report with key recommendations. This report offers actionable advice,

aligning technical, legal, practical, and strategic considerations to support informed decision-making. Key recommendations might include implementing robust environmental management plans to comply with enhanced obligations and leveraging the heightened environmental focus to strengthen market positioning. By presenting a cohesive and comprehensive analysis, the system equips stakeholders with the necessary information to navigate the contractual changes effectively.

This case study highlights the effectiveness of the multi-agent LLM framework in handling complex contractual analyses. The coordinated agents deliver accurate, comprehensive insights that support informed decision-making, maintaining professional standards throughout the analysis process. By simplifying the explanation while retaining core arguments, the narrative becomes clearer and more focused, enhancing the reader's understanding of the system's capabilities.

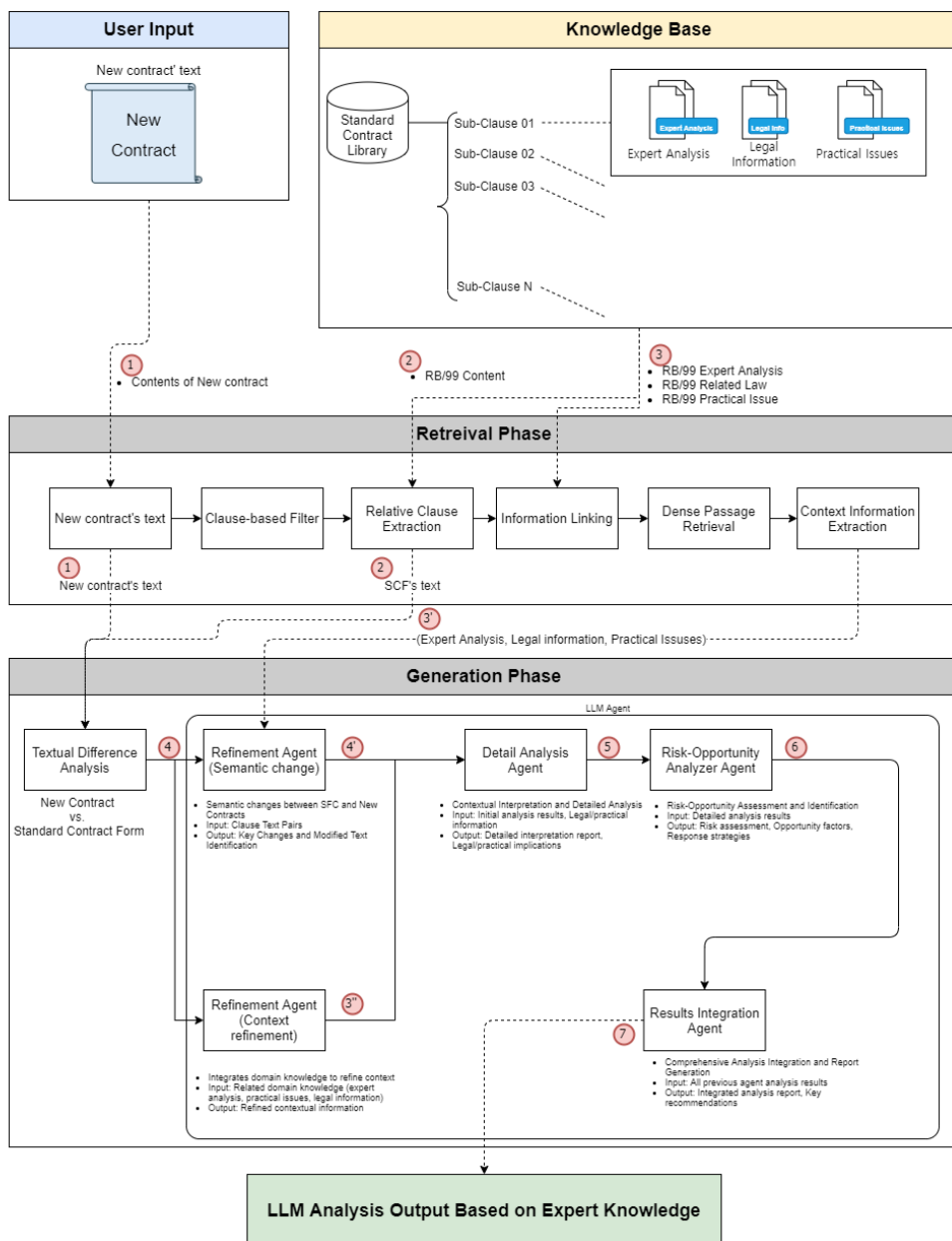


Figure 11. Multi-stage Knowledge Integration and Analysis Process

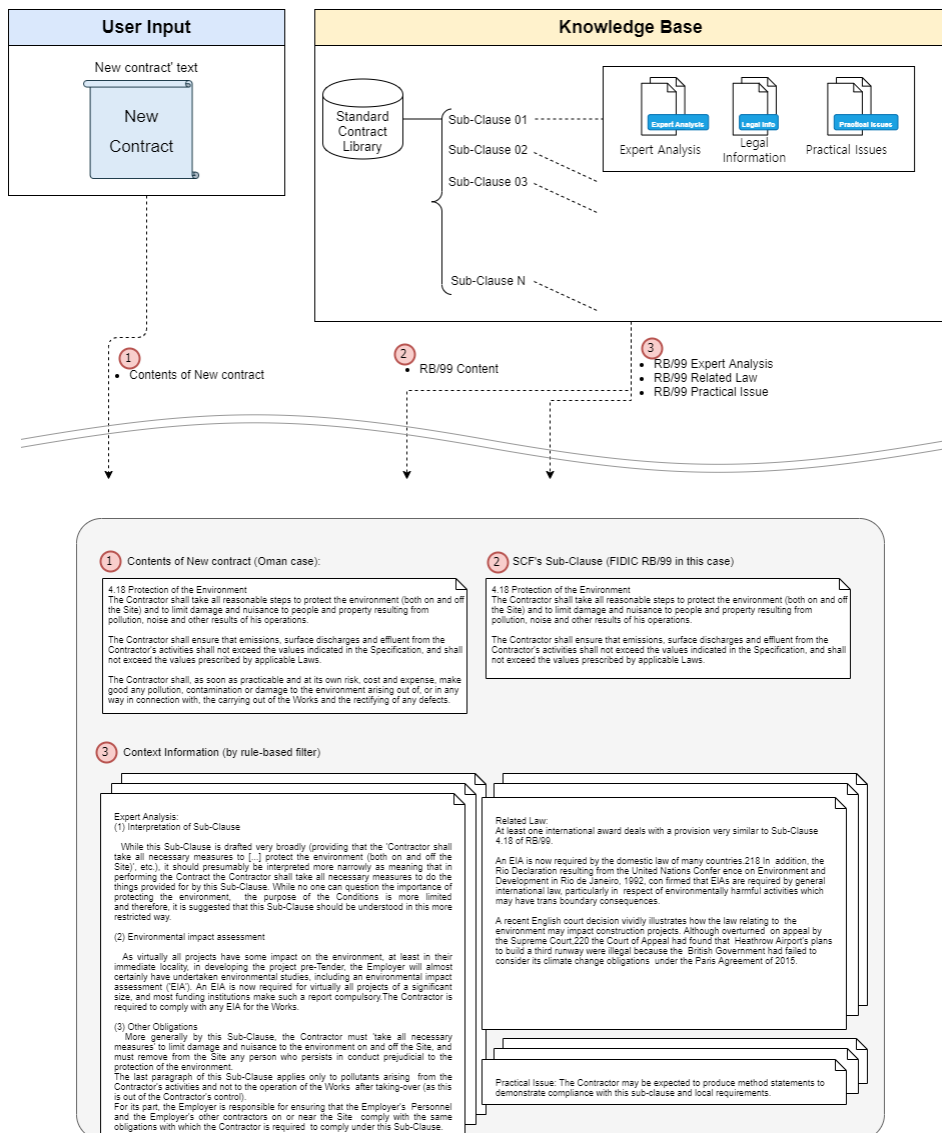


Figure 12. Initial Processing and Knowledge Base Architecture

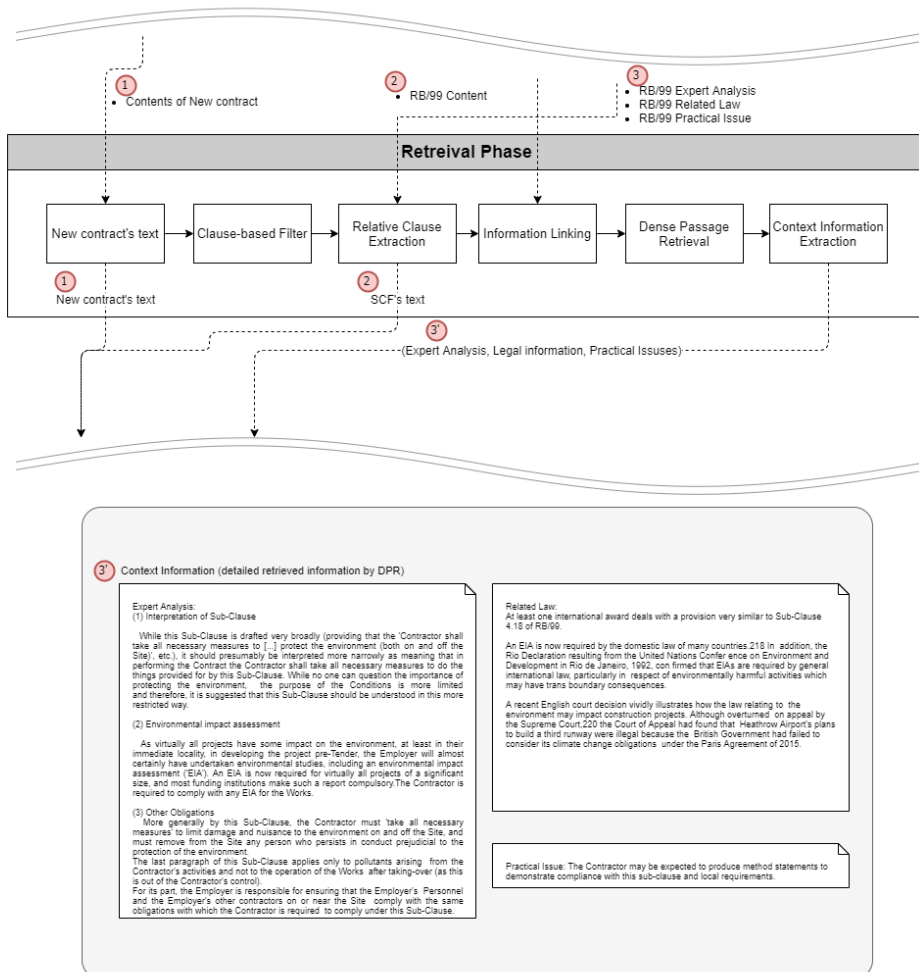


Figure 13. Result of Context Information Extraction

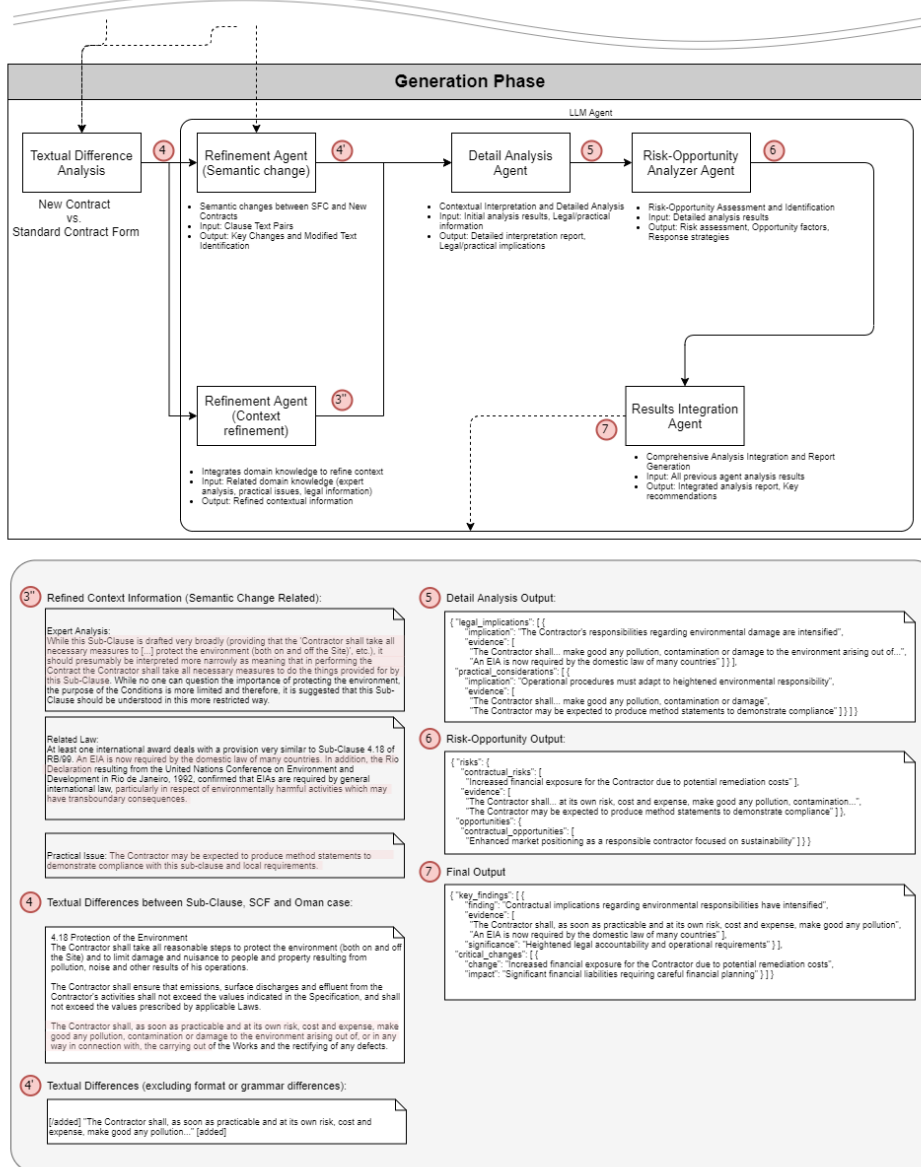


Figure 14. Results of Each Agent Generation

4.3 Expert Evaluation

4.3.1 Evaluation Methodology

The expert evaluation assessed the system's performance, usability, and practical applicability, focusing on the roles of the specialized LLM agents. Thirteen domain experts from construction, engineering, legal advisory, and client organizations participated. They were given direct access to the prototype system to interact with it firsthand, with detailed usage instructions provided as outlined in Appendix III.

In addition to using the system, one-hour interviews were conducted with each expert to gather in-depth feedback. The interviews were structured using a prepared evaluation form included in Appendix I. Responses were collected on various aspects, including clause types, performance metrics, and productivity enhancements. This combination of direct system interaction and structured interviews ensured a comprehensive evaluation of the system's capabilities and limitations from professional perspectives. The respondents' distribution is summarized in Table 11, detailing their industry roles, age groups, and experience levels.

Table 11. Expert information

Category	Subcategory	Number of Respondents
Job Role	Contractor	7
	Client	2
	Consultant	3
Age Group	30s	8
	40s	2
	50s	1
	60s	1
Experience Level	5–10 years	8
	10–20 years	1
	Over 20 years	3

The evaluation was structured to provide both quantitative and qualitative assessments of the system's performance across several dimensions. The experts evaluated the system based on specific criteria related to clause types, performance metrics, and productivity enhancements.

(1) Clause Types Assessed

Table 12. Clause Types Assessed

Clause Type	Description
Modified Clauses (Major Changes)	Significant alterations to standard clauses.
Modified Clauses (Minor Changes)	Minor modifications to standard clauses.
Additional Clauses	New clauses added to the contract.
Missing Clauses	Standard clauses omitted from the contract.

(2) Evaluation Categories and Metrics

The evaluation focused on three main categories, each with specific metrics, as outlined in Table 13.

Table 13. Evaluation Categories and Metrics

Category	Evaluation Metric	Description
Reference Information (Search Performance)	Relevance of Reference Information	The extent to which the retrieved information is pertinent to the clauses being reviewed.
	Sufficiency of Reference Information	The adequacy of the information provided, considering both quantitative and qualitative aspects.
Clause Analysis Performance	Clause Interpretation	Accuracy and clarity in interpreting contractual language.
	Practical Issues Analysis	Identification and analysis of practical implications for contract execution.
	Legal Analysis	Depth and correctness of legal implications identified.
	Risk & Opportunity Assessment	Ability to identify risks and opportunities arising from contractual changes, and the appropriateness of the analysis.
Productivity	Feasibility of Practical Implementation	The potential for the system to be adopted in real-world contract review processes.
	Work Efficiency Improvement Effect	The extent to which the system enhances productivity by reducing analysis time and automating tasks.
	Degree of Decision Support	How well the system supports decision-making processes.

(3) Scoring Criteria

Experts rated each metric on a scale of 1 to 5, corresponding to different levels of expertise, as detailed in Table 14.

Table 14. Scoring Criteria

Score	Experience Level	Description
1 Point	Novice (0–2 years of experience)	- Lacks ability to acquire and utilize relevant information. - Unable to identify risks effectively.
2 Points	Equivalent to 2–4 years of experience	- Basic ability to acquire and utilize information. - Can identify and analyze obvious risks.
3 Points	Intermediate (4–7 years of experience)	- Capable of reviewing contract risks for individual clauses.
4 Points	Equivalent to 7–10 years of experience	- Able to review contract risks from an integrated perspective.
5 Points	Expert (Over 10 years of experience)	- Performs expert-level contract clause reviews.

4.3.2 Performance Assessment and System Capabilities

The experts provided both quantitative ratings and qualitative feedback on the system's performance. The quantitative results are

summarized in Table 15, and the qualitative insights are presented in Table 16.

The evaluation demonstrated that the system performed well in several areas:

- **Efficiency and Productivity:** The metrics under Productivity, specifically Practical Issue Identification and Time Savings, received relatively high average scores of 3.54 and 3.85, respectively. This indicates that the system significantly reduces contract review time and aids practitioners in identifying practical issues efficiently. Experts highlighted that the system enhances work efficiency and has the potential to alleviate the workload for practitioners.
- **Search Performance:** The Relevance of Retrieved Information and Sufficiency of Information metrics received average scores of 3.08 and 3.31, respectively. These scores suggest that the system is moderately effective in retrieving relevant clauses and providing sufficient information for basic contract analysis. The Textual Difference Analysis Agent was particularly

effective in identifying significant variations between contracts.

However, the evaluation also identified areas where the system needs improvement:

- **Analysis Performance:** The metrics related to Interpretation, Practical Issues, Legal Analysis, and Risk & Opportunity Analysis received lower average scores, ranging from 2.08 to 2.77. These scores indicate limitations in the system's ability to perform in-depth analysis and provide detailed legal insights. The Semantic Analysis Agent and Context Analysis Agent need enhancements to better handle complex legal interpretations and practical considerations.

4.3.3 Analytical Capability and Limitations

The lower scores in the Analysis Performance category highlight specific challenges:

(1) Interpretation (2.77)

- Corresponding Experience Level: Approximately 2–4 years of experience (Equivalent to 2 Points).
- Analysis: With an average score close to 3, the system's ability to accurately interpret contractual language and its implications is between basic and intermediate levels. It aligns with the capabilities of a practitioner with 2–4 years of experience. This indicates that the system can handle standard clauses and perform basic interpretations but struggles with nuanced language that requires deeper understanding typically possessed by more experienced professionals (those with over 4 years of experience).

(2) Practical Issues Analysis (Average Score: 2.23) and Legal Analysis (Average Score: 2.08):

- Corresponding Experience Level: Around 2–4 years of experience (Equivalent to 2 Points).
- Analysis: The system shows limitations in identifying practical implications and providing detailed legal analysis, performing at a level comparable to a practitioner with 2–4 years of experience. It possesses a basic ability to acquire

and utilize information and can identify and analyze obvious risks but lacks the depth and specificity expected from more experienced professionals. Experts noted the absence of information on applicable laws and precedents, which are critical for comprehensive legal analysis.

(3) Risk & Opportunity Assessment (Average Score: 2.38):

- Corresponding Experience Level: Approximately 2–4 years of experience (Equivalent to 2 Points).
- Analysis: The system's difficulty in distinguishing between risks and opportunities is reflected in its average score, indicating performance at a junior level. The unclear analysis criteria hinder its reliability in supporting decision-making processes that depend on accurate risk assessments. This level of performance suggests that the system can identify obvious risks but may not effectively evaluate more complex or nuanced situations, a capability typically developed by practitioners with more than 4 years of experience.

These limitations suggest that while the system effectively aids in

preliminary reviews and enhances productivity for tasks requiring basic analysis, it currently operates at an experience level equivalent to that of a junior professional with 2–4 years of experience. To serve as a comprehensive tool for in-depth contract analysis comparable to that performed by intermediate to expert-level practitioners (those with over 4 years of experience), the system requires further development.

Improving the analytical capabilities of the Semantic Analysis Agent and Context Analysis Agent is essential to elevate the system's performance to match higher experience levels. By enhancing the system to perform at a level equivalent to practitioners with 4–7 years of experience (Intermediate level, 3 Points) or above, it can better handle complex legal interpretations, provide detailed legal analyses, and support more accurate risk assessments. This advancement would enable the system to address nuanced contractual language and intricate legal implications, fulfilling the needs of more advanced contract analysis tasks in the construction industry.

4.3.4 Implementation Potential and Future Directions

Despite the identified challenges, experts expressed optimism regarding the system's potential for practical implementation. The higher

scores in Practical Issue Identification and Time Savings, along with positive qualitative feedback, reinforce the system's usefulness in streamlining contract review processes.

The experts provided insights on future development priorities:

- **Enhancing Analytical Depth:** Improving the Semantic Analysis Agent and Context Analysis Agent to deliver more accurate and detailed interpretations and legal analyses. Incorporating information on applicable laws, precedents, and critical risk items such as cash flow is essential.
- **Clarifying Risk Analysis Criteria:** Refining the Risk & Opportunity Assessment Agent to establish clear criteria and methodologies for risk assessment, enabling it to distinguish effectively between risks and opportunities.
- **Technical Enhancements:** Addressing processing speed limitations to handle larger datasets and urgent tasks more efficiently. Enhancing the user interface for better clarity and usability will contribute to wider adoption and effectiveness.

The expert evaluation provided valuable insights into the system's current capabilities and areas needing improvement. The system demonstrates significant potential in enhancing efficiency and productivity in contract reviews. However, to serve as a comprehensive tool for detailed legal analysis, further development and refinement of the LLM agents are necessary. By addressing the identified challenges, the system can be enhanced to meet the complex demands of contract analysis in the construction industry.

Table 15. Expert evaluation results

Category	Metric	Average	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13
Search Performance	Relevance of Retrieved Information	3.08	4	3	3	3	4	2	3	2	3	3	3	3	4
	Sufficiency of Information	3.31	4	3	5	3	4	2	3	3	3	3	3	3	4
Analysis Performance	Interpretation	2.77	4	2	4	2	4	2	2	2	2	2	2	4	4
	Practical Issues	2.23	4	2	3	2	4	1	1	1	1	1	2	3	4
	Legal Analysis	2.08	4	2	3	1	3	1	2	2	1	2	1	2	3
	Risk & Opportunity Analysis	2.38	4	2	4	2	3	1	2	2	1	2	2	2	4
Productivity	Practical Issue Identification	3.54	5	2	3	5	4	3	3	3	3	3	3	4	5
	Time Savings	3.85	4	2	4	4	4	3	4	4	4	4	4	4	5

Table 16. Expert Feedback Summary

Category	Positive Feedback	Challenges
Efficiency and Productivity	- Significantly reduces contract review time, enhancing work efficiency. - Potential to alleviate workload for practitioners. - Assists junior professionals in understanding and approaching contracts. - Can be utilized as an auxiliary tool for contract management by quickly comparing with company standard contracts.	- Processing speed is insufficient for large datasets or urgent tasks. - Manual intervention is required for detailed analysis and validation. - Final judgment still needs human expertise, raising applicability concerns.
Analytical Capability	- Provides excellent relevance and quantity of information. - Effectively summarizes issues related to contract clauses. - Anticipated improvements in work efficiency and reduction of legal costs.	- Lacks information on applicable laws and precedents. - Risk analysis criteria are unclear, making it difficult to distinguish between risks and opportunities. - Insufficient analysis of legal issues with low accuracy and specificity. - Lacks emphasis on important risk items such as cash flow.

4.4 Summary

This chapter presented the CORA-RAG system, a multi-agent Large Language Model framework developed to automate and enhance the contract review process in the construction industry. By employing agents—such as those for textual difference analysis, semantic analysis, context evaluation, and risk assessment—the system facilitates efficient and comprehensive contract analysis.

The pipeline architecture integrates these agents to ensure coherent progression through analysis stages, enhancing the accuracy and reliability of the outputs. Validation mechanisms involving domain experts are incorporated to maintain quality.

The system was evaluated both through automated metrics and an expert assessment involving thirteen professionals from contractors, clients, and consulting firms. Results indicated that the system effectively improves efficiency and productivity, significantly reducing contract review time and assisting practitioners, particularly less experienced ones, in understanding complex contracts.

However, the evaluation revealed that the system's analytical capabilities currently align with the proficiency level of junior professionals

with 2–4 years of experience. This was evidenced by average scores ranging from 2.08 to 2.77 in areas such as clause interpretation, practical issues analysis, legal analysis, and risk-opportunity assessment. The system demonstrates basic abilities to acquire and utilize information and can identify and analyze obvious risks but struggles with nuanced interpretations and in-depth legal analysis that require expertise comparable to more seasoned practitioners (those with over 4 years of experience).

Challenges were identified in handling complex legal interpretations, providing detailed legal insights, and distinguishing effectively between risks and opportunities. Limitations in processing speed and the need for manual intervention in detailed analyses were also noted.

Experts expressed optimism about the system's potential for practical implementation, suggesting future development should focus on:

- **Enhancing the Analytical Capabilities:** Improving the agents to perform at higher proficiency levels, equivalent to practitioners with 4–7 years of experience or more. This includes providing more detailed legal insights, accurately interpreting nuanced contractual language, and offering comprehensive analyses.

- **Clarifying Risk Analysis Criteria:** Refining the risk assessment methodologies to establish clear criteria, enabling the system to distinguish more effectively between risks and opportunities, and support decision-making processes.
- **Improving Technical Performance:** Enhancing processing speed to handle larger datasets and urgent tasks efficiently, and improving user interface usability for better user experience.

By addressing these areas, the CORA-RAG system can evolve into a comprehensive tool capable of meeting the complex demands of real-world contract analysis in the construction industry. Advancing the system's capabilities to match higher experience levels will enable it to handle intricate legal scenarios and provide in-depth analyses, supporting professionals in making better-informed decisions and managing risks more effectively.

5. Discussion

5.1 Technical Performance Analysis

5.1.1 System Architecture and Integration

The CORA-RAG system tackles challenges in automating construction contract reviews by integrating refinement and generation components within a multi-agent LLM framework. agent LLMs focus on specific tasks like legal analysis, technical evaluation, and risk assessment, applying appropriate expertise to enhance accuracy and relevance. This specialization allows the system to handle complex contract clauses effectively.

Experimental results show that the CORA-RAG system improves upon base LLMs and standard RAG approaches in automated metrics such as refinement quality and response generation. However, expert evaluations reveal that the system's real-world effectiveness aligns with the performance level of junior professionals with 2–4 years of experience. While the system enhances certain technical aspects, further development is needed to achieve higher proficiency levels in complex contract analysis.

5.1.2 Multi-stage Generation Process

The generation component employs a multi-stage process where agent LLMs refine contextual data and integrate expert knowledge from various domains. Agents handle tasks like differential analysis, and risk-opportunity assessment, accurately identifying differences in contract clauses and ensuring coherence across technical and legal domains.

Currently, the system provides insights encompassing technical and practical aspects but lacks depth in legal analysis and nuanced interpretations. Its performance is comparable to practitioners with 2–4 years of experience, indicating the need for further refinement to generate comprehensive legal insights.

5.2 Practical Implications and Limitations

5.2.1 System Benefits and Applications

The CORA-RAG system automates the initial stages of contract review, accelerating the identification of relevant clauses by quickly processing large volumes of text and highlighting deviations. This reduces manual effort and time, enhancing the overall efficiency of contract management processes—particularly valuable in the construction industry,

where contractual documents are extensive and complex.

By efficiently refining information and identifying clause changes, the system assists practitioners, especially those with less experience, in navigating complex contracts. It enhances transparency by documenting each agent's steps—such as sources consulted and reasoning applied—building trust in the automated review process.

5.2.2 Current Limitations and Challenges

Despite its strengths, the CORA-RAG system currently operates at an analytical level equivalent to junior professionals with 2–4 years of experience. It exhibits limitations in handling highly complex clauses, nuanced legal scenarios, and distinguishing between risks and opportunities, affecting its reliability in supporting decision-making processes that depend on accurate risk assessments and in-depth legal analyses.

Improvements are needed in processing speed and user interface to facilitate seamless integration into professional workflows. Enhancing agent LLM performance is essential to elevate the system's analytical capabilities to match those of more experienced practitioners.

While automated metrics show positive results, discrepancies with

expert evaluations highlight the limitations of relying solely on these measures for assessing real-world effectiveness. Manual expert evaluation remains crucial for ensuring high-quality contract analysis and guiding system refinement to better align with professional standards.

Future development should focus on enhancing the analytical capabilities of the agents to perform at higher proficiency levels, equivalent to practitioners with 4–7 years of experience or more. This includes improving the ability to interpret nuanced contractual language, providing detailed legal analyses, and effectively distinguishing between risks and opportunities. Advancements in these areas are vital for evolving the CORA-RAG system into a comprehensive tool capable of meeting the complex demands of real-world contract analysis in the construction industry.

6. Conclusion

6.1 Research Summary and Achievements

This research addressed three major challenges in automating construction contract review by developing a Structured Knowledge-Integrated RAG Pipeline that integrates domain expertise with LLMs through a multi-agent architecture.

Firstly, to tackle the challenge of organizing domain expertise, a structured knowledge framework was developed. This framework systematically integrates expert analyses, legal information, and practical considerations into a knowledge base aligned with SCFs. By mapping domain knowledge to specific SCF clauses and employing classification-based filtering, the system effectively reduces irrelevant data early in the retrieval process, enhancing accuracy and efficiency compared to methods relying solely on dense and sparse searches without prior filtering.

Secondly, to overcome the challenge of maintaining accuracy with complex and extensive contract documents, a two-stage refinement process within a multi-agent LLM system was implemented. Agents are assigned to tasks such as semantic analysis and context analysis, allowing the system to handle intricate contractual conditions. Enhancing context during refinement

and integrating hybrid search techniques significantly improved information accuracy. Experimental validation showed that Context Recall reached 0.969 and Faithfulness scored 0.893, indicating substantial improvements over basic RAG approaches and demonstrating the system's high accuracy with complex contracts.

Thirdly, to address the challenge of ensuring reliability in LLM-generated content, reliable generation processes were established through a multi-agent LLM framework. The system employs a multi-stage refinement pipeline where agent LLMs collaborate to align content generation with domain expertise. The reference-based generation process maintains explicit connections to source materials by citing specific clauses and referencing the knowledge base during content generation. This approach enhances traceability and trustworthiness, ensuring that all generated analyses can be directly linked back to authoritative sources. The system achieved an average score of 0.828 ± 0.087 , indicating consistent outputs, though Context Precision of 0.625 and Answer Correctness of 0.750 suggest opportunities for refining retrieval precision and response generation.

6.2 Research Contributions

This research advances the field of construction contract management through three significant contributions:

1. **Structured Integration of Domain Knowledge:** A framework was introduced for integrating domain expertise into automated contract review, transforming the organization and application of construction contract knowledge. By systematically organizing expert analyses, legal information, and practical considerations into a structured knowledge base using ontologies and metadata tagging, the system can access and apply expert interpretations, legal precedents, and practical guidelines during analysis. This enables comprehensive and reliable assessments by drawing on a rich repository of domain-specific knowledge.

2. **Enhanced Retrieval Accuracy through Classification-Based Filtering:** Remarkable improvements in retrieval accuracy were achieved by implementing classification-based filtering combined with hybrid dense-sparse retrieval techniques. By categorizing and filtering the knowledge base according to relevant SCF clauses, the system reduces irrelevant data early in the process, improving retrieval efficiency and accuracy. Notable performance gains were observed, with Context Recall improving by 52.6% and Faithfulness by 48.3%, setting new standards for automated document analysis, especially in the context of complex construction contracts.

3. Reliable Generation Processes through Multi-Agent LLM

Framework: A multi-agent LLM framework was developed to enhance the reliability and accuracy of content generation. By employing a multi-stage refinement pipeline with agent LLMs, the system aligns content generation with domain expertise. The reference-based generation process maintains explicit connections to source materials by citing specific clauses and referencing the knowledge base during content generation. This approach enhances traceability and trustworthiness, ensuring that all generated analyses can be directly linked back to authoritative sources.

6.3 Research Limitations

Despite its advancements, the CORA-RAG system has several limitations:

1. **Retrieval Precision and Answer Correctness:** While Context Recall and Faithfulness scores are high, a Context Precision of 0.625 and Answer Correctness of 0.750 indicate that retrieval sometimes includes less relevant information, and responses may not always be fully accurate. Refining retrieval algorithms and improving answer generation processes are necessary to enhance precision and correctness.

2. **Dependence on Knowledge Base Scope:** The system's effectiveness heavily relies on the comprehensiveness of its knowledge base concerning standard form contracts. This reliance may limit its applicability to contracts outside of this scope, such as customized or significantly modified contracts, reducing its practicality in varied contractual scenarios.

3. **Complexity of Multi-Agent LLM Framework:** The multi-agent architecture, while enhancing reliability, introduces complexity in agent coordination and communication. Challenges may arise in ensuring seamless collaboration among agents, potentially affecting processing speed and efficiency. Further research is needed to optimize agent interactions and manage the computational overhead associated with the multi-agent system.

6.4 Future Research

To enhance the CORA-RAG system and address its limitations, future research should prioritize the following areas:

1. **Enhancing Agent LLM Capabilities:** Improve the training and collaboration of agent LLMs to handle intricate clauses and subtle contractual nuances more effectively. Advanced training on diverse legal scenarios and optimization of agent communication protocols will enhance

their interpretative abilities and processing efficiency.

2. Improving Retrieval Precision and Response Generation: Develop more sophisticated retrieval algorithms and refine the generation processes to increase Context Precision and Answer Correctness. Optimizing the integration of the retrieval and generation components will improve information flow and relevance. Implementing advanced validation mechanisms can ensure the accuracy of generated content.

3. Expanding the Knowledge Base: Extend the knowledge base beyond standard form contracts to include tailored and non-standard contracts. This expansion will enhance the system's practical applicability across a broader range of contractual scenarios, making it more versatile and valuable in the construction industry.

Advancements in these areas will require establishing interdisciplinary teams comprising researchers, legal professionals, and construction industry practitioners. Regular workshops, collaborative development sessions, and pilot projects can facilitate the exchange of expertise, ensuring that system enhancements align with practical requirements and industry needs. By focusing on enhancing agent LLM capacities, refining retrieval and generation processes, and expanding the knowledge base, the CORA-RAG system can further bridge the gap between

technical capabilities and real-world utility. Although the system represents a significant step forward in automating construction contract processes, ongoing refinement—driven by both academic rigor and industry demands—will be essential for transforming contract management practices in the construction sector.

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Appendix I. Expert Evaluation Form

계약조항 검토 자동화 시스템 전문가 평가

안녕하십니까,

본 전문가 평가에 참여해 주셔서 진심으로 감사드립니다.

연구 목적

- 계약서 기본검토 단계에서 소요되는 과도한 시간과 인력부족 문제를 해결하기 위해 인공지능 기술을 활용한 계약조항 검토 자동화 시스템을 개발하였습니다
- 전문가들의 의견을 통해 시스템의 성능을 평가하고자 합니다

평가 개요

- 소요시간: 약 20 분
- 응답대상: 건설계약 전문가 약 30 명
- 평가내용: 구체적 사례를 기반으로 시스템의 검색성능, 분석성능, 실무 적용성 등을 평가합니다.
- 해당 시스템은 FIDIC Red book 99 년판과 Oman 의 General condition 의 차이점에 중점을 두며, **시공자의 관점**에서 분석하도록 구축되었습니다.
- 평가는 수정된 조항(major & minor), 추가된 조항, 삭제된 조항 네가지 유형에 대해 각각 최소한 개씩 성능평가를 요청드립니다.
- 예시
 - 수정된 조항(major change): 16.2 Termination by Contractor (Oman)
 - 수정된 조항(minor change): 4.19 Utilities (Oman)
 - 추가된 조항: 17 Anti-Bribery and Corruption (Oman)
 - 삭제된 조항: 13.8 Adjustments for Changes in Cost (FIDIC RB/99)

개인정보 보호

- 설문 및 평가결과는 연구 목적으로만 사용되며, 개인정보는 익명으로 처리됩니다

문의사항

- 연구책임자: 김의왕 박사과정
- 소속: 중앙대학교
- 연락처: kew1013@cau.ac.kr, 010-8970-5061

본 평가에 참여해 주셔서 진심으로 감사드립니다.

2024. . .

※ 응답자 정보

구 분	분류정보 입력란
① 응답자의 소속유형 (중복응답 가능)	시공(), 엔지니어링(), 감리/건설사업관리(), 법무(), 컨설턴트(), 발주처(), 기타()
② 응답자의 전문분야 (중복응답 가능)	건설(), 법률(), 기타()
③ 응답자의 경력기간	20년≤경력(), 10년≤경력<20년(), 5년≤경력<10년(), 경력<5년()
④ 응답자의 연령	① 20대() ② 30대() ③ 40대() ④ 50대() ⑤ 60대 이상()

※ 평가항목 설명

조항 유형	평가부문	평가 지표	상세
•수정된 조항(major) •수정된 조항(minor) •추가 조항 •누락 조항	1.참조정보 검색성능	• 참조정보의 연관성 • 참조정보의 충분성	• 양적/질적 측면
	2.조항 분석성능	• 조항 해석 • 실무이슈 분석 • 법률 분석 • Risk & Opportunity	• 분석의 적정성
	3.생산성	• 실무도입 가능성 • 업무효율 개선 효과	• 분석 소요시간, 자동화 수준 • 의사결정 지원 정도

※ 평가점수 기준

평가점수	1점	2점	3점	4점	5점
기준	초급 (0-2 년차)	- (2-4 년차)	중급 (4-7 년차)	- (7-10 년차)	전문가 (10 년차 이상)
역량	• 관련 정보 획득/활용 능력 부족 • 리스크 식별 능력 부족	• 기본적인 정보 획득/활용 • 표면적으로 드러난 리스크 식별/분석 가능	• 개별 계약조항 별 계약리스크 검토 가능	• 통합적 관점에서 계약리스크 검토 가능	• 전문가 수준 계약조항 검토

1. 참조정보 검색 성능

1.1 정보의 연관성 (질적 측면)

초급자	-	중급자	-	전문가
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): 예시: 검색된 정보의 품질이 좋지 않음 (+ 문제점 발견된 조항)

1.2 정보의 충분성 (양적)

초급자	-	중급자	-	전문가
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): 예시: '~' 조항의 분석에 필요한 필수정보가 부족함 (+ 문제점 발견된 조항)

2. 계약조항 분석 성능

2.1 해석 기능(Interpretation)

초급자	-	중급자	-	전문가
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): 예시: 유형별(major change, minor change, 추가 조항, 누락 조항) 성능차이 존재(+ 문제점 발견된 조항)

2.2 실무이슈 분석 기능(Practical Issues)

초급	-	중급	-	전문가
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): 예시: 조항 및 맥락 이해도 부족(+ 문제점 발견된 조항)

2.3 법률분석 기능(Legal Information)

초급	-	중급	-	전문가
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): _____

2.4 Risk & Opportunity 분석 기능

초급	-	중급	-	전문가
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): _____

3. 생산성

3.1 실무도입 가능성

부정	약간 부정	중립	약간 긍정	긍정
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): 예시: '-/오류, 정확도 측면에 문제가 있어 실무적용에 어려움이 있음

3.2 업무 효율 개선효과

부정	약간 부정	중립	약간 긍정	긍정
☐ 1점	☐ 2점	☐ 3점	☐ 4점	☐ 5점

- 특이사항(선택): 예시: 기초분석 단계에서 '-/' 시간(-%) 단축 가능

※ 기타 의견

(선택사항) 시스템에 관한 의견이 있으시면 자유롭게 작성 부탁드립니다.

예시: 추가적으로 필요한 기능, 실무 적용시 예상되는 문제점

· Oman 계약조건에서 변경된 조항

· 수정된 조항(minor)	· 수정된 조항(major)
1.13 Compliance with Laws & Procedures	1.2 Interpretation
2.1 Right of Access to the Site	1.3 Communications
2.4 Employer's Claims	1.4 Law and Language
4.8 Quality Assurance	1.5 Priority of Documents
4.9 Site Data	1.6 Contract Agreement
4.12 Unforeseeable Physical Conditions	1.7 Assignment
4.15 Access Rout	1.8 Care and Supply of Documents
4.17 Contractor's Equipment	1.9 Delayed Drawings or Instructions
4.19 Utilities	1.14 Joint and Several Liability
5.3 Payments to nominated Subcontractor	2.2 Permits, Licences or Approvals
5.4 Evidence of Payments	3.1 Engineer's Duties and Authority
7.3 Samples	3.4 Replacement of the Engineer
7.7 Rejection	4.2 Performance Bond
7.8 Remedial Work	4.4 Subcontractors
7.10 Ownership of Plant and Materials	4.6 Co-operation
7.11 Royalties	4.18 Protection of the Environment
8.1 Commencement of Works	4.22 Security of the Site
8.9 Suspension of Work	4.23 Contractor's Operations on Site
8.10 Consequences of Suspension	6.1 Engagement of Staff and Labour
8.11 Payment for Plant and Materials in Event of Suspension	6.4 Labour Laws
9.1 Contractor's Obligations	6.6 Facilities for Staff and Labour
11.1 Completion of Outstanding Work and Remedying Defects	6.7 Health and Safety
11.2 Cost of Remedying Defects	6.10 Records of Contractor's Personnel and Equipment
11.3 Extension of Defects Notification Period	7.2 Manner of Execution
11.10 Performance Certificate	7.4 Inspection
12.1 Estimated Quantities	8.2 Time for Completion
12.4 Method of Measurement	8.3 Program
13.1 Right to Vary	8.4 Extension of Time for Completion
13.5 Daywork	8.7 Delay Penalties
14.2 Advance Payment	8.13 Resumption of Work
14.12 Statement at Completion	9.4 Failure to Pass Tests on Completion
14.13 Application for Final Payment Certificate	10.1 Taking Over the Works and Sections
14.14 Discharge	10.3 Interference with Tests on Completion
14.15 Issue of Final Payment Certificate	11.5 Removal of Defective Work
14.16 Cessation of Employer's Liability	11.7 Contractor to Search
15.3 Valuation at Date of Termination	11.9 Right of Access
16.3 Cessation of Work and Removal of Contractor's Equipment	11.11 Unfilled Obligations

19.1 Indemnities	12.5 Evaluation
19.2 Contractor's Care of the Works	13.2 Value Engineering
19.3 Employer's Risks for Loss and Damage	13.3 Variation Procedure
19.4 Consequences of Employer's Risks	13.4 Provisional Sums
18 Intellectual and Industrial Property Rights	14.3 Application for Interim Payment Certificates
20.1 General Requirements for Insurances	14.4 Schedule of Payments
20.2 Insurance for Works & Equipment	14.7 Payment
20.3 Insurance against Injury to Persons& Damage to Property	14.9 Delayed Payment
20.4 Insurance for Contractor's Personnel	14.10 Payment of Retention Money
21.1 Definition of Force Majeure	14.8 Currency of Payment
21.2 Notice of Force Majeure	15.5 Employer's Termination for Convenience
21.3 Duty to Minimise Delay	16.1 Contractor's Entitlement to Suspend Work
21.4 Consequences of Force Majeure	16.2 Termination by Contractor
21.5 Force Majeure Affecting Subcontractor	16.4 Payment on Termination
21.6 Optional Termination and Payment	19.5 Limitation of Liability
21.7 Release from Performance under the Law	22.2 Amicable Settlement
22.3 Litigation	22.4 Arbitration

· RB/99 대비 추가된 조항

1.15 Confidentiality
4.11 Errors in Computing Accepted Contract Amount
6.11 Expatriate Staff
7.1 Quality of Materials and Goods
7.6 Cost of Tests
7.9 Delivery of Plant
8.8 Engineering Services
11.8 Latent Defects
12.2 Actual and Correct Quantities
14.11 Cash Flow
17 Anti-Bribery and Corruption

· RB/99 대비 삭제된 조항(FIDIC의 조항번호 및 타이틀입니다)

2.4 Employer's Financial Arrangements
4.8 Safety Procedures
11.8 Contractor to Search
12.4 Omissions
deleted condition
13.8 Adjustments for Changes in Cost
20.2 Appointment of the Dispute Adjudication Board
20.3 Failure to Agree Dispute Adjudication Board
20.4 Obtaining Dispute Adjudication Board's Decision
20.7 Failure to Comply with Dispute Adjudication Board's Decision
20.8 Expiry of Dispute Adjudication Board's Appointment

Appendix II. Used prompts in retrieval phase

(a) Base LLM

Please return ONLY the answer to the following question, excluding the question itself:

Question: [Question]

Please provide a concise and clear answer that directly addresses the question, without including the question in your response.

Example Format:

Q: How do contractor rules for fossils and archaeology change in reports, like notice and claims?

A: Contractor rules for fossils and archaeology in reports require the Contractor to notify the Engineer upon discovering findings, describe the finding in their notice, and follow the claims procedure outlined in Sub-Clause 20.2. The Engineer must inspect the finding promptly, and the Contractor can request an extension of time and additional payment for delays or costs arising from following the Engineer's instructions.

Remember: Return ONLY your answer without repeating the question.

(b) Perplexity

System Message:

You are an AI assistant specialized in FIDIC Red Book 1999 analysis.

User Message:

Please analyze the following FIDIC Red Book 1999 Sub-Clause:

Sub-Clause Title: [Title]

Question: [Question]

Follow these steps:

1. Research Sub-Clause focusing on:

- Original text
- Analysis and interpretation
- Practical implementation issues
- Legal implications and cases
- Areas for improvement

2. Provide response in this format:

[ANSWER START]

Direct answer to the question based on research

[ANSWER END]

[CONTEXT START]

||Contents of Sub-Clause(RB/99): [Original text]

||Analysis: [Interpretation]

||Practical issues: [Implementation challenges]

||Related Law: [Legal context]

||Improvements: [Suggested improvements]

[CONTEXT END]

(c) Standard RAG

Please return ONLY the answer to the following question, excluding the question itself:

Question: [Question]

Context:

Contents of Sub-Clause(RB/99): [Original clause text]

Main Changes: [Changes from RB/99]

Analysis: [Detailed analysis]

Related Law: [Legal context]

Improvements: [Suggested improvements]

Please provide a concise and clear answer that directly addresses the question, without including the question in your response.

Example Format:

Q: How do contractor rules for fossils and archaeology change in reports, like notice and claims?

A: Contractor rules for fossils and archaeology in reports require the Contractor to notify the Engineer upon discovering findings, describe the finding in their notice, and follow the claims procedure outlined in Sub-Clause 20.2. The Engineer must inspect the finding promptly, and the Contractor can request an extension of time and additional payment for delays or costs arising from following the Engineer's instructions.

Remember: Return ONLY your answer without repeating the question.

(d) Proposed retrieval method

System Message:

You are an AI assistant specialized in FIDIC Red Book 1999 analysis.

User Input:

Sub-Clause Title: [Title]

Question: [Question]

Context: [Retrieved context from hybrid search]

Analysis Steps:

1. Analyze the provided context focusing on:

- Original Sub-Clause text
- Sub-Clause analysis
- Implementation issues
- Legal implications and cases
- Areas for improvement

Response Format:

[ANSWER START]

Direct answer to the question based on the retrieved context

[ANSWER END]

[CONTEXT START]

||Contents of Sub-Clause(RB/99): [Original text from context]

||Analysis: [Analysis based on context]

||Practical issues: [Implementation challenges]

||Related Law: [Legal implications]

||Improvements: [Areas for improvement]

[CONTEXT END]

Appendix III. Used prompts in generation phase

(a) Refiner Agent Prompt (Semantic change)

Analyze the differences and filter out non-substantial changes.

Ignore:

- Formatting changes
- Grammar or punctuation changes
- Word order changes that don't alter meaning
- Simple rewording with same meaning

Focus on substantive changes affecting:

- Rights and obligations
- Legal implications
- Risk allocation

Return Format:

```
{
  "status": "success",
  "changes": {
    "substantive_differences": [
      {
        "red99_clause": "원문 조항 텍스트",
        "oman_clause": "변경된 조항 텍스트"
      }
    ]
  }
}
```

(b) Refiner Agent Prompt (Context refinement)

Analyze and refine the text spans related to the differences found in `text_diff`.

Focus on the following source contents:

- red99Analysis: Contains analysis sections
- red99RelatedLaw: Contains legal references
- red99Practicalissue: Contains practical implications

Return Format:

```
{  
  "status": "success",  
  "refined_spans": {  
    "analysis": [관련 계약해석 내용],  
    "legal": [관련 법률 내용],  
    "practical": [관련 실무 내용]  
  }  
}
```

(c) Detail Analyst Prompt

For each substantive difference:

1. Analyze legal implications:

- Connect to exact clause changes

2. Examine practical considerations:

- Link to operational impacts
- Implementation challenges

3. Evaluate risk factors:

- Direct consequences
- Potential cascading effects

Return Format:

```
{
  "status": "success",
  "analysis": {
    "legal_implications": [...],
    "practical_considerations": [...],
    "risk_factors": [...],
    "interrelationships": [...]
  }
}
```

(d) Risk/Opportunity Analyst Prompt

Analyze opportunities focusing on:

1. Identify opportunities from:

- Legal compliance advantages
- Operational efficiency gains
- Market positioning benefits

2. Support with:

- Direct evidence
- Market context
- Quantifiable benefits

Return Format:

```
{
  "status": "success",
  "assessment": {
    "risks": {
      "contractual_risks": [...],
      "evidence": [...],
      "reasoning": "..."
    },
    "opportunities": {
      "contractual_opportunities": [...],
      "evidence": [...],
      "reasoning": "..."
    }
  }
}
```

(e) Aggregator Agent Prompt

Provide detailed comparative analysis:

1. Specific Key Findings:

- Identify exact differences in contractual implications

2. Critical Changes:

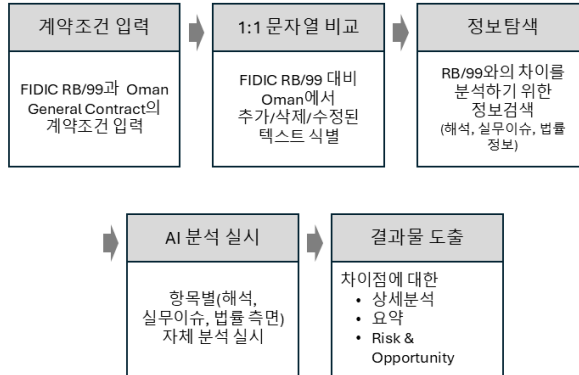
- List concrete differences in responsibilities
- Reference specific findings
- Explain significance
- Show reasoning process

Return Format:

```
{
  "status": "success",
  "summary": {
    "key_findings": [...],
    "critical_changes": [...]
  }
}
```

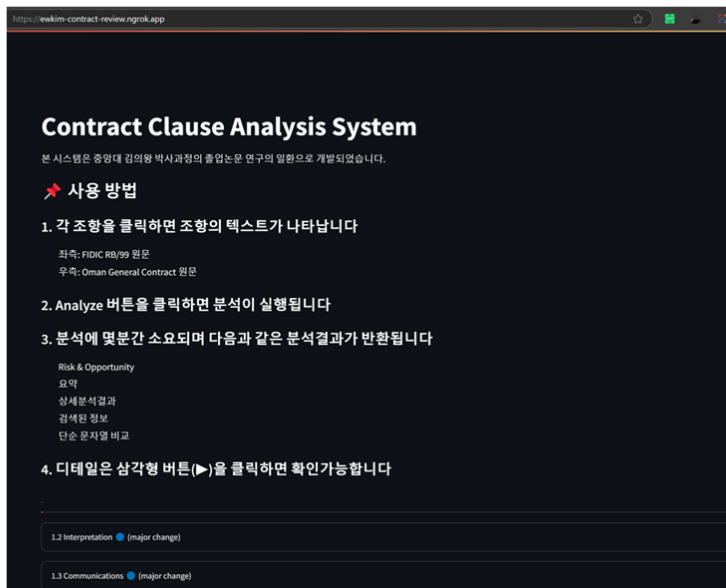
Appendix IV. Guide for Expert Assessment

시스템 작동 절차



1. Url에 접속하면 다음과 같은 화면이 나옵니다

<https://ewkim-contract-review.ngrok.app/>



2. 아래로 스크롤하면 Oman 표준계약서의 조항이 목록으로 나타납니다

해당 조항이 FIDIC RB/99 기준으로 어떻게 수정되었는지를 나타냅니다 (major/minor change, deleted, added 중)

1.2 Interpretation	● (major change)	▼
1.3 Communications	● (major change)	▼
1.4 Law and Language	● (major change)	▼
1.5 Priority of Documents	● (major change)	▼
1.6 Contract Agreement	● (major change)	▼
1.7 Assignment	● (major change)	▼
1.8 Care and Supply of Documents	● (major change)	▼
1.9 Delayed Drawings or Instructions	● (major change)	▼
1.13 Compliance with Laws & Procedures	● (minor change)	▼
1.14 Joint and Several Liability	● (major change)	▼

3. 조항을 클릭하면 다음과 같이 RB/99와 Oman의 텍스트가 나타납니다

1.14 Joint and Several Liability ● (major change)

FIDIC RB/99 - Clause 13

Text

1.14 Joint and Several Liability
If the Contractor constitutes (under applicable Laws) a joint venture, consortium or other unincorporated grouping of two or more persons:
(a) these persons shall be deemed to be jointly and severally liable to the Employer for the performance of the Contract;
(b) these persons shall notify the Employer of their leader who shall have authority to bind the Contractor and each of these persons; and
(c) the Contractor shall not alter its composition or legal status without the prior consent of the

Analyze 13

Oman General Contract - Clause 13

Text

(b) these Persons shall notify the Employer of their leader who shall have authority to bind the Contractor and each of these Persons; and
(c) the Contractor shall not alter its composition or legal status without the prior consent of the Employer.
(d) each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.

클릭

Oman의 텍스트는 편의상 미리 세팅해두었습니다.
오른쪽의 텍스트는 수정 가능하며, 원하시는 조건으로 변경후에 분석도 가능합니다

기능1. 문자열 비교

FIDIC RB/99 대비 제거된 부분은 빨간색, 추가된 부분은 파란색으로 강조되어 나타납니다

아래의 케이스에서는 “each of such Persons shall ... payment.”가 추가되었습니다

1.14 Joint and Several Liability

If the Contractor constitutes a joint venture, consortium or other unincorporated grouping of two or more Persons:

(a) these Persons shall be deemed to be jointly and severally liable to the Employer for the performance of the Contract;

(b) these Persons shall notify the Employer of their leader who shall have authority to bind the Contractor and each of these Persons; and

(c) the Contractor shall not alter its composition or legal status without the prior consent of the Employer.

Of each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.

Analysis completed!

필요정보 검색결과	상세분석결과
단순 문자열 분석:	[...]
Interpretation:	Detailed Analysis: Interpretation
[...]	[...]
Practical:	Detailed Analysis: Practical Issues
[...]	[...]
Legal:	Detailed Analysis: Legal Information
[...]	[...]

* 시각화 기능에서는 내부적으로는 정상작동됨에도 불구하고 색깔, 빗금처리예 예러가 발생할 때가 있습니다. 내부적인 분석에는 지장이 없으니 양해부탁드립니다

기능2. 정보검색(왼쪽)

- 상각형 버튼을 클릭하면 아래와 같이 상세한 결과물 확인이 가능합니다
- 필요정보 검색결과: RB/99와 Oman 계약조항의 차이를 분석하기 위해 검색된 레퍼런스입니다

필요정보 검색결과

단순 문자열 분석:

[...]

Interpretation:

[...]

Practical:

[...]

Legal:

[...]

해석관련 검색된 정보

실무이슈 관련 검색된 정보

법률 관련 검색된 정보

어떤 레퍼런스가 검색되었는지 상세한 내용을 확인할 수 있습니다

```

Legal:
{
  "legal_analysis": {
    "key_changes": {
      0: {
        "differences": {
          "added": {
            0:
            "each of such Persons shall recognise any payment required
            to be paid by the Employer to the Contractor under the
            Contract as being fully paid and satisfied if such payment
            is made to the account of the Contractor in accordance with
            the Contract, and such Persons shall not have any rights or
            claims against the Employer in respect of the manner of
            division of such payment."
          }
          "deleted": {}
          "words_changed": {}
        }
      }
    }
    "extracted_legal":
    "(10) Related Law: The law governing the contract creating the JV and
    governing the JV undertaking is likely, as a practical matter, to be
    the law of the country of at least one of the members of the JV,
    perhaps the leader of the JV. The General Conditions do not regulate
    this matter. However, the JV and the JV undertaking must comply with
    Sub-Clauses 1.14 and 1.14.1 as interpreted by the law governing the
    Contract provided for in Sub-Clause 1.4. There may be no
    inconsistencies.377
  }
  377. International legal principles relating to joint and several
  obligations are to be found in Chapter 11 [Plurality of Obligors and
  Obligees], UNIDROIT Principles."
  }
}
                    
```

추가된 문자열 “each of such Persons shall ... payment.”에 대한 법률정보 검색결과입니다

기능3. 상세분석(오른쪽)

- RB/99와 Oman 계약조항의 차이에 대해서 해석 / 실무이슈 / 법률 측면에서 조항을 분석한 결과입니다

상세분석결과

▶ {...}

Detailed Analysis: Interpretation

▶ {...}

Detailed Analysis: Practical Issues

▶ {...}

Detailed Analysis: Legal Information

▶ {...}

➔

Detailed Analysis: Interpretation

```

{
  0: {
    "topic": "Joint and Several Liability in Joint Ventures"
    "analysis":
      "The Oman clause reaffirms the principle of joint and several liability among joint venture members, ensuring that the Employer can hold any single member accountable for the entire performance of the contract, should others default. This provision increases the security for the Employer compared to a scenario where members are only severally liable."
    "impact":
      "This provision enhances the Employer's position in contract execution, as it guarantees that they can seek redress from any member of the joint venture without needing to pursue each member individually. It simplifies the process of enforcement and reduces the risk of non-performance."
    "considerations":
      "It is crucial for joint venture members to understand the implications of joint and several liability, as it increases their potential exposure to risk. They should also ensure that the JV leader is clearly identified and authorized to act on behalf of the group to prevent complications in communication with the Employer."
  }
}
          
```

Detailed Analysis: Practical Issues

```

{
  0: {...}
  1: {...}
  2: {...}
  3: {...}
}
          
```

Detailed Analysis: Legal Information

```

{
          
```

기능4. 요약

- 앞서 도출된 상세분석결과를 간략하게 요약한 결과입니다

텍스트가 추가/삭제/변경인지를 나타내는 부분

추가된 텍스트(Oman)

해석에 대한 요약

레퍼런스

실무이슈에 대한 요약

법률정보에 대한 요약

요약

```

{
  "Key_Changes": [
    0: {
      "type": "added"
      "source_text":
        "each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or respect of the manner of division of such payment."
      "description":
        "This change clarifies the recognition of payments made to the Contractor's account and limits any claims against the Employer regarding..."
      "impact": {
        "interpretation": {
          "summary_content":
            "Members of the JV cannot claim against the Employer regarding the division of payments, which protects the Employer but may members."
          "reference": {...}
        }
        "practical": {
          "summary_content":
            "Establish clear internal agreements on payment distribution among JV members to prevent disputes and ensure all members are expectations."
          "reference": {...}
        }
        "legal": {
          "summary_content": "Members of the joint venture are jointly and severally liable to the Employer, increasing their risk exposure."
          "reference": {...}
        }
        "significance": "high"
        "criteria":
          "This change was selected as important due to its potential to trigger internal disputes among JV members and its role in safeguarding interests."
      }
    }
  ]
}
          
```

기능5: Risk & Opportunity 도출

- Risk and Opportunity: 최종적으로 계약조건의 차이점으로 인해 어떤 Risk/Opportunity가 식별되었는지를 나타냅니다

Risk and Opportunity Analysis							
Sub-Clause No.	Title	Text Difference	Type	Description	Importance	Reasoning	Evidence
0.1.14	Joint and Several Liability	[added]each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.[added]	Risk	Members may dispute payment allocation, leading to internal conflicts within the JV.	Medium	["Interpretation": "The added clause limits JV members' rights to claim against the Employer's payment distribution, which could strain member relations," "Practical": "Clear internal agreements are vital to avoid disputes over payment distribution, which could hinder project execution," "Legal": "The clause enhances the Employer's position by protecting them from claims by JV members regarding payment distribution."]	["Interpretation": "The clause constrains JV members from claiming against the Employer, thus preserving the Employer's interests but potentially leading to disputes among JV members regarding payment division," "Practical": "Establishing clear internal agreements on payment distribution will be essential to prevent disputes and align financial expectations among members," "Legal": "The legal implication of this clause means that JV members are limited in recourse against the Employer, which may lead to conflicts among themselves regarding payment allocations."]
1.1.14	Joint and Several Liability	[added]each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.[added]	Opportunity	The clause simplifies the payment process for the Employer, enhancing contract performance assurance.	High	["Interpretation": "The clause clarifies payment recognition, which provides stability in payment processes," "Practical": "This clarity can lead to smoother operations and reduced disputes over payment, benefiting the overall project execution," "Legal": "The legal framework ensures that payments made to the Contractor's account are considered settled, reducing potential claims against the Employer."]	["Interpretation": "By ensuring that payments to the Contractor's account are recognized as satisfied, the clause protects the Employer and minimizes complexities in payment processes," "Practical": "With this clear payment recognition, the Employer can operate more efficiently without worrying about claims from JV members regarding payment distribution," "Legal": "Legally, this provision strengthens the Employer's position, providing a clear basis for payment execution without subsequent claims from JV members."]

5. 전체 분석결과는 함께 송부해드린 엑셀파일을 통해서 확인하실 수 있습니다

Overall Analysis Results							
Sub-Clause No.	Title	Text Difference	Type	Description	Importance	Reasoning	Evidence
0.1.14	Joint and Several Liability	[added]each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.[added]	Risk	Members may dispute payment allocation, leading to internal conflicts within the JV.	Medium	["Interpretation": "The added clause limits JV members' rights to claim against the Employer's payment distribution, which could strain member relations," "Practical": "Clear internal agreements are vital to avoid disputes over payment distribution, which could hinder project execution," "Legal": "The clause enhances the Employer's position by protecting them from claims by JV members regarding payment distribution."]	["Interpretation": "The clause constrains JV members from claiming against the Employer, thus preserving the Employer's interests but potentially leading to disputes among JV members regarding payment division," "Practical": "Establishing clear internal agreements on payment distribution will be essential to prevent disputes and align financial expectations among members," "Legal": "The legal implication of this clause means that JV members are limited in recourse against the Employer, which may lead to conflicts among themselves regarding payment allocations."]
1.1.14	Joint and Several Liability	[added]each of such Persons shall recognise any payment required to be paid by the Employer to the Contractor under the Contract as being fully paid and satisfied if such payment is made to the account of the Contractor in accordance with the Contract, and such Persons shall not have any rights or claims against the Employer in respect of the manner of division of such payment.[added]	Opportunity	The clause simplifies the payment process for the Employer, enhancing contract performance assurance.	High	["Interpretation": "The clause clarifies payment recognition, which provides stability in payment processes," "Practical": "This clarity can lead to smoother operations and reduced disputes over payment, benefiting the overall project execution," "Legal": "The legal framework ensures that payments made to the Contractor's account are considered settled, reducing potential claims against the Employer."]	["Interpretation": "By ensuring that payments to the Contractor's account are recognized as satisfied, the clause protects the Employer and minimizes complexities in payment processes," "Practical": "With this clear payment recognition, the Employer can operate more efficiently without worrying about claims from JV members regarding payment distribution," "Legal": "Legally, this provision strengthens the Employer's position, providing a clear basis for payment execution without subsequent claims from JV members."]

국문초록

김의왕
스마트시티학과
스마트계획·건설정보시스템전공
중앙대학교 대학원

건설계약 검토는 기술과 법률 분야의 전문지식을 요구하지만, 현업에서는 전문인력 부족으로 많은 어려움을 겪고 있다. 대규모언어모델(LLM)이 이러한 문제해결에 잠재력을 보이고 있으나, 복잡한 계약에서 도메인지식을 효과적으로 통합하고 활용하여 신뢰할 수 있는 결과물을 생성하는 데는 여전히 한계가 있다. 이러한 문제를 해결하기 위해 본 연구에서는 CORA-RAG 시스템을 제안한다.

CORA-RAG는 2,000개 이상의 텍스트 세그먼트를 포함하는 지식 베이스를 기반으로 계약조항의 해석이 가능하다. 해당 시스템은 멀티 에이전트 접근 방식을 채택하여 의미 분석, 컨텍스트 분석, 위험 평가 등 다양한 측면에서 분석을 수행한다. 분석은 차이점 식별, 컨텍스트 정제,

실무 및 법률적 측면의 처리, 상세한 리스크 및 기회 평가 단계를 거치며, 이는 에이전트 간 협업을 통해 종합적인 생성 프레임워크 내에서 이루어진다.

오만의 General Conditions를 이용한 성능평가를 실시하였으며 제안된 방법은 기존 RAG 접근방식 대비 Context Recall 52.6%, Faithfulness 48.3% 증가 등 유의미한 개선을 보였다. 분류기반 필터링과 하이브리드 검색방법의 도입으로 초기 과정에서 불필요한 정보를 효과적으로 제거했으며, 멀티 에이전트 LLM 프레임워크를 통해 분석결과의 신뢰성과 정확성을 향상시켰다. 건설산업 분야의 13명의 전문가 평가 결과, 제안된 시스템은 경력 2~4년차 주니어 전문가 수준의 분석능력을 갖는것으로 나타났다. 전문가들은 해당 시스템이 기본적인 계약분석을 신속하게 수행하고 검토시간을 단축하여 업무 효율성을 향상시키는 데 유용하며, 특히 경험이 적은 실무자들이 복잡한 계약을 이해 및 분석하는 데 큰 도움이 될 것으로 평가되었다.

본 연구의 주요 공헌은 SCF 기반의 구조화된 지식 프레임워크

개발, 분류기반 필터링과 하이브리드 검색 방법을 통한 정보 검색 정확도 향상, 그리고 멀티 에이전트 LLM 프레임워크를 통한 컨텍스트 인식 다단계 답변 생성 프로세스 개발이다. 이러한 실용적인 접근을 통해 CORA-RAG는 건설계약 분야에서 증가하는 자동화된 솔루션 수요에 대응하는 신뢰성 높고 효율적인 도구로 활용이 가능하다.

핵심어: AI기반 계약서 검토, 계약관리, 검색증강생성(RAG), 대규모 언어모델(LLM), 지식베이스